

UNIT-II

DATA ENCRYPTION TECHNIQUES AND STANDARDS

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Unit II	Data Encryption Techniques And Standards	(08 Hours)
Introduction, Encryption Methods: Symmetric, Asymmetric, Cryptography, Substitution Ciphers. Transposition Ciphers, Stenography applications and limitations, Block Ciphers and methods of operations, Feistel Cipher, Data Encryption Standard (DES), Triple DES, Weak Keys in DES Algorithms, Advance Encryption Standard (AES).		

CRYPTOGRAPHY

- Cryptography is technique of **securing information** and **communications** through **use of codes** so that only those person for whom the information is intended can understand it and process it.
- Thus preventing unauthorized access to information.
- The prefix “crypt” means “hidden” and suffix “graph” means “writing”.



TECHNIQUES USED FOR CRYPTOGRAPHY:

- In today's age of computers **cryptography** is often **associated with** the process
- Where plain text is converted to cipher text which decode by receiver only , this process is known as **encryption**.
- The process of conversion of cipher text to plain text this is known as **decryption**.

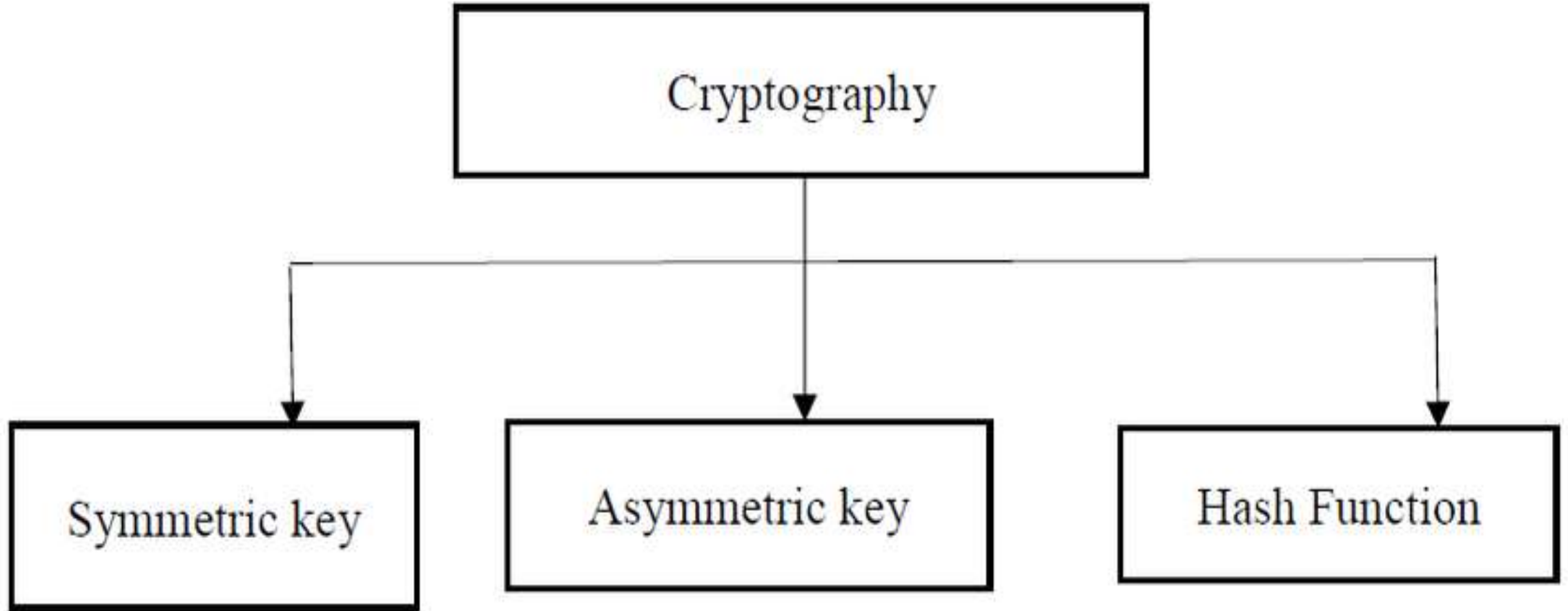
What are Plaintext and Ciphertext? How do they interact?

- **Plaintext** can refer to anything which **humans can understand and/or relate to**.
- This may be as simple as **English sentences**, a script, or Java code.
- If you can make sense of what is written, then it is in **plaintext**.
- **Cipher text**, or encrypted text, is a **series of randomized letters and numbers** which humans cannot make any sense of.
- An **encryption algorithm** takes in a plaintext message, runs the algorithm on the plaintext, and produces a cipher text.
- The cipher text can be reversed through the process of **decryption**, to produce the original plaintext.

FEATURES OF CRYPTOGRAPHY ARE AS FOLLOWS:

- **Confidentiality:**
information can only be **accessed by the person** for whom it is intended and no other person except him can access it.
- **Integrity:**
information **cannot be modified in storage** or transition between sender and intended receiver without any addition to information being detected.
- **Non-repudiation:**
the creator/sender of information cannot **deny his or her intention** to send information at later stage.
- **Authentication:**
the identities of sender and receiver **are confirmed**. As well as destination/origin of information is confirmed.

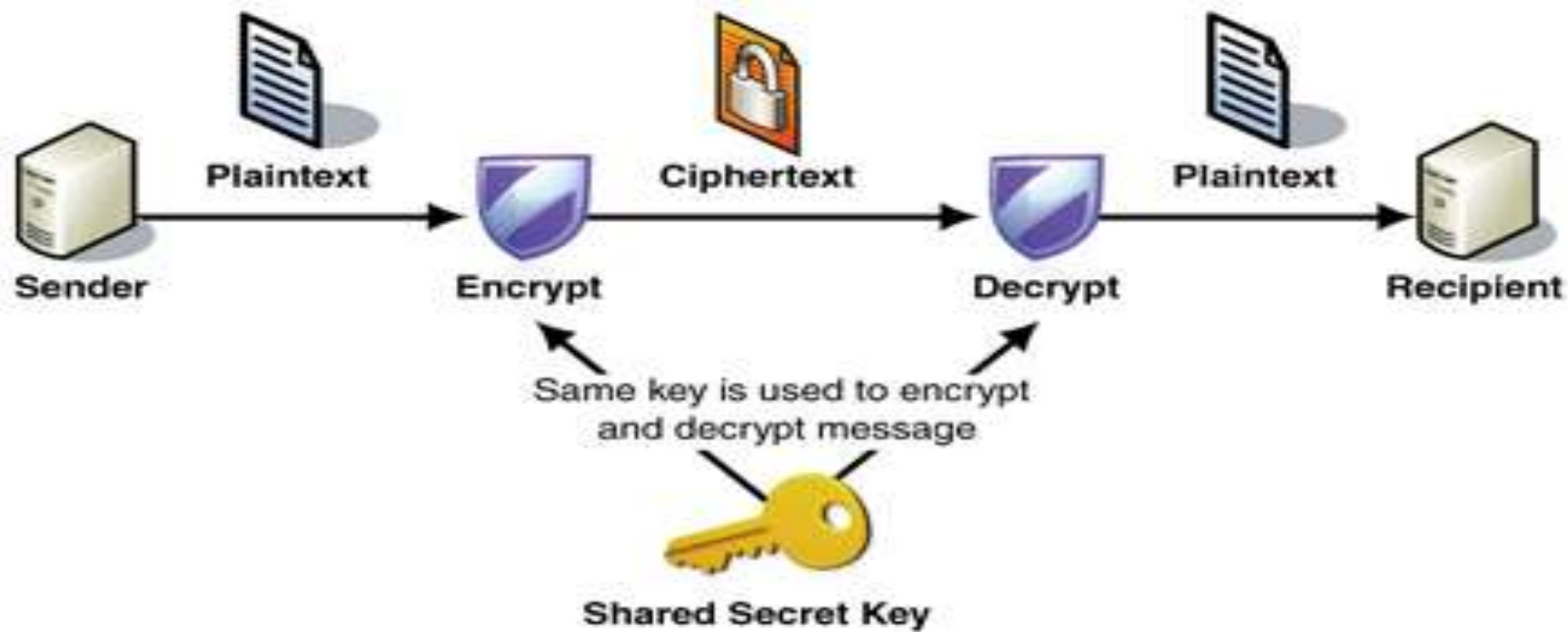
TYPES OF CRYPTOGRAPHY



TYPES OF CRYPTOGRAPHY:

1.Symmetric key cryptography(Secret key cryptography)(Single/Private key encryption)

- It is system where the sender and receiver uses a **single common key** to encrypt and decrypt messages.
- Symmetric key systems are **faster and simpler**
- But the problem is that sender and receiver have to somehow **exchange key in a secure manner**.
- Example: **data encryption system(des)**.



Why to use symmetric key encryption?

- It is relatively **inexpensive** to produce a **strong key** for these ciphers.
- The keys tend to be much **smaller** for the level of protection they afford.
- The **algorithms** are relatively **inexpensive** to process.
- The **drawback** :
- to secret-key ciphers is in **exchanging the secret key** because any exchange must retain the privacy of the key.

2. ASYMMETRIC KEY CRYPTOGRAPHY (public-key encryption) (asymmetric encryption)

- Under this system a pair of keys is used to encrypt and decrypt information.
- A public key is used for encryption and a private key is used for decryption.
- Public key and private key are different.
- Even if the public key is known by everyone the intended receiver can only decode it because he alone knows the private key.

Asymmetric Encryption



Why to use Asymmetric key encryption?

- the encrypted data is not understood by an intruder.
- **Non repudiation**, which prevents:
 - The sender of the data from claiming, at a later date, that the data was never sent
 - The data from being altered.
- RSA (Rivest-Shamir-Adelman) public key pairs can be any size. Typical sizes today are 1024 and 2048 bits.

3.HASH FUNCTIONS:

- There is no usage of **any key** in this algorithm.
- A hash value with **fixed length is calculated as per the plain text** which makes it impossible for contents of plain text to be recovered.
- Many **operating systems** use hash functions to **encrypt passwords**.



Cryptographic Hash Functions

[,krip-tə-'gra-fik 'hash 'fən(k)-shənz]

A mathematical function used in cryptography which typically take inputs of variable lengths to return outputs of a fixed length.

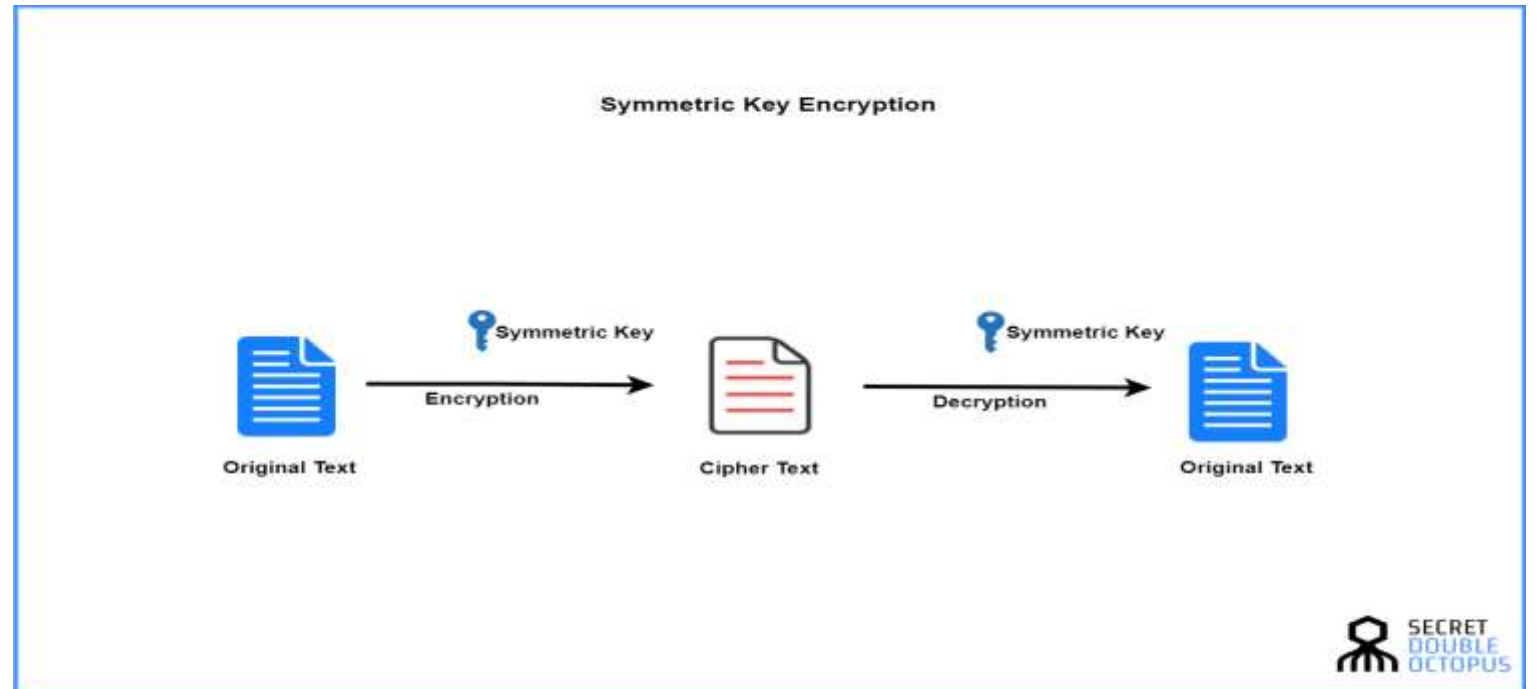
CLASSICAL ENCRYPTION TECHNIQUES:

- In the classical **cryptography** the original data i.e., The plain text is transformed into the **encoded format** i.e. Cipher text so that we can transmit this data through insecure communication channels.
- A data string which known as **key** is used to control the transformation of the data from plain text to cipher text.
- This **arrangement helps to keep data safe** as it required the key for extracting the original information from the cipher text.
- **Without the key no one can read the data.**
- In this technique it is assumed that the **only authorized receiver has the key.**

CLASSICAL CRYPTOGRAPHY HAS TWO TYPES OF TECHNIQUES:

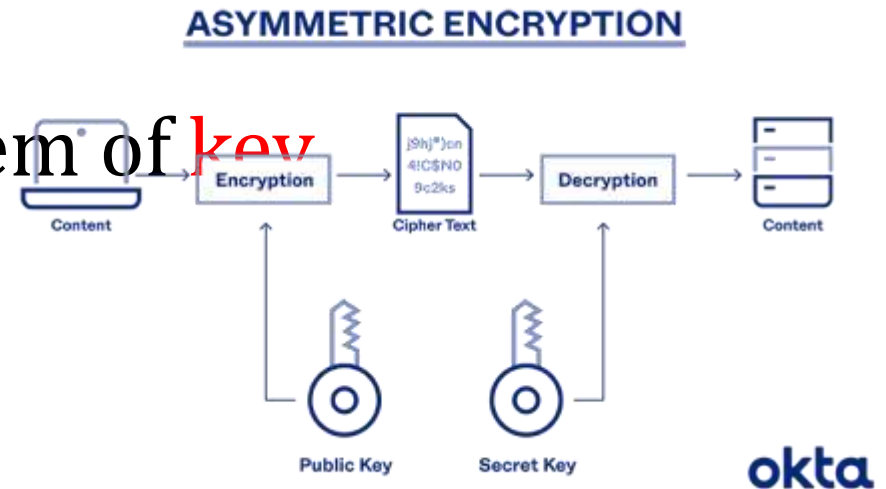
Symmetric Cryptography:

- In the symmetric cryptography a **single key is used** for encrypting and decryption the data.
- This encryption key is **private key**.
- This is the limitation of this encryption technique that **this private key must be distributed only among the authorized sender and receiver.**



ASYMMETRIC CRYPTOGRAPHY:

- In the asymmetric cryptography **a pair of key**, i.e., **Public key and Private key** is used for encryption and decryption.
- A sender can use its **public key to encrypt the data** and on receiver end receiver can decrypt the data by using its **private key**.
- This technique overcomes the problem of **key distribution**.



ADVANTAGES AND DISADVANTAGES OF CLASSICAL CRYPTOGRAPHY:

Advantages

- While employing the **one-time pad**(information-theoretically secure), it is unbreakable.
- It is easy to do **manually**, no computer required.
- It protects the plain text from casual **snooping**(keep watch on what person is typing.).

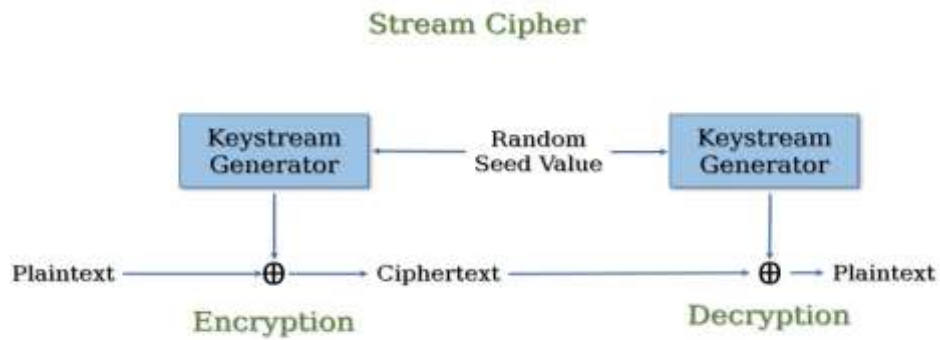
Disadvantages

- While employing the one-time pad, it is cumbersome and requires a personal meet up **to exchange the pads**.
- If not employing the OTP, anyone who is even remotely interested in knowing what you wrote and knows about cryptography will be able **to break the encryption**.

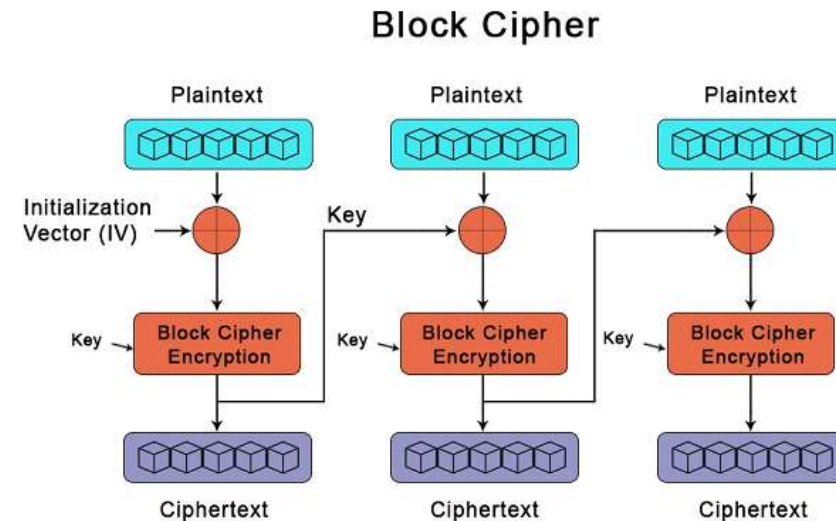
BLOCK CIPHER AND STREAM CIPHER

- **Block cipher** and **stream cipher** belongs to the **symmetric key cipher**.
- These two are the methods used for converting the **plain text** into **cipher text**.
- A block cipher converts the plain text into cipher text by **taking plain text's block at a time**

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S. NO	Block Cipher	Stream Cipher
1.	Block Cipher Converts the plain text into cipher text by taking plain text's block at a time.	Stream Cipher Converts the plain text into cipher text by taking 1 byte of plain text at a time.
2.	Block cipher uses either 64 bits or more than 64 bits.	While stream cipher uses 8 bits.
3.	The complexity of block cipher is simple.	While stream cipher is more complex.
4.	Block cipher Uses confusion as well as diffusion.	While stream cipher uses only confusion.
5.	In block cipher, reverse encrypted text is hard.	While in-stream cipher, reverse encrypted text is easy.
6.	The algorithm modes : ECB (Electronic Code Book) and CBC (Cipher Block Chaining).	The algorithm modes : CFB (Cipher Feedback) and OFB (Output Feedback).
7.	Block cipher works on transposition techniques like rail-fence technique, columnar transposition technique, etc.	While stream cipher works on substitution techniques like Caesar cipher, polygram substitution cipher, etc.
8.	Block cipher is slow	While stream cipher is fast

ATTACKS ON SYMMETRIC KEY CRYPTOGRAPHY:

- Typically, the objective of attacking an encryption system is to recover the **key** in use rather than simply **to recover the plaintext** of a single cipher text.
- There are two general approaches to attacking a conventional encryption scheme.
- **Cryptanalysis:**
- **Brute-force attack**

- **Cryptanalysis:** cryptanalytic attacks rely on the nature of the algorithm plus knowledge of the general characteristics of the plaintext or even some sample plaintext-cipher text pairs.
- This type of attack exploits the characteristics of the algorithm to attempt to deduce a specific plaintext or to deduce the key being used.
- **Brute-force attack:** the attacker tries every possible key on a piece of cipher text until an intelligible translation into plaintext is obtained.
- On average, half of all possible keys must be tried to achieve success.
- A brute force attack is a hacking method that uses trial and error to crack passwords, [login credentials](#), and encryption keys.
- It is a simple yet reliable tactic for gaining unauthorized access to individual accounts and organizations' systems and networks.

Substitution (cipher) techniques

- The two basic building blocks of all encryption Techniques are **substitution** and **transposition**.
- A substitution technique
- Definition:
- Letters of **plaintext** are replaced by **other letters** or **by numbers or symbols** .
- if the plaintext is viewed as a **sequence of bits**,
- then substitution involves **replacing plaintext bit patterns** with **cipher text bit patterns**.

Substitution Techniques

1. Caesar Cipher
2. Monoalphabetic Cipher
3. Polyalphabetic Cipher
4. Playfair
5. Hill
6. One time Pad

1. CAESAR CIPHER:

- The earliest known, and the simplest, use of a substitution cipher was by **julius caesar**.
- The caesar cipher involves **replacing each letter of the alphabet with the letter standing **three** places further down the alphabet.**
- For example, **(encryption)**

Plain: meet me after the toga party

cipher: PHHW PH DIWHU WKH WRJD SDUWB

- As follows:
- **plain:** a b c d e f g h i j k l m n o p q r s t u v w x y z
- **cipher:** D E F G H I J K L M N O P Q R S T U V W X Y Z A B C

- **Decryption** is just as easy, by using an offset of -3.

- For example, **(decryption)**

Cipher: PHHW PH DIWHU WKH WRJD SDUWB

Plain: meet me after the toga party

- If it is known that a given cipher text is a **caesar cipher**, then a **brute-force cryptanalysis** is easily performed: simply try all the **25 possible keys**.
- **Disadvantage:**
 1. The encryption and decryption **algorithms are known**.
 2. There are only **25 keys** to try.
 3. The language of the plaintext is known and **easily recognizable**

2. Monoalphabetic Cipher

- A **Caesar shift cipher/mono-alphabetic cipher (simple substitution cipher)** is a substitution cipher
- where each letter of the **plain text is replaced with another letter** of the alphabet.
- It uses a **fixed key** which consist of the **26 letters** of a “shuffled alphabet”.
- all “A” letters in the plain text will be encoded to an “M”.
- This type of cipher is a form of **symmetric encryption** as the **same key** can be used to both **encrypt** and **decrypt** a message.

Plain text alphabet:	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Cipher text alphabet (key):	M	U	A	L	V	O	Z	K	R	N	J	X	Q	D	F	S	H	P	E	B	C	T	I

3. Polyalphabetic cipher

- Poly alphabetic cipher is far more secure than a mono alphabetic cipher.
- As mono-alphabetic cipher maps a plain text symbol or alphabet to a cipher text symbol and uses the same cipher text symbol wherever that plain text occurs in the message.
- But poly alphabetic cipher, each time replaces the plain text with the different cipher text.

- with only 25 possible keys, the caesar cipher is far from secure.
- a dramatic **increase in the key space** can be achieved by allowing an arbitrary substitution. before proceeding, we define the term **permutation**. a permutation of a finite set of elements s is an ordered sequence of all the elements of s , with each element appearing exactly once.
- for example, if $s = \{a, b, c\}$,
- there are **six permutations** of s : abc, acb, bac, bca, cab, cba
- in general, there are **$n!$** permutations of a set of n elements, because the first element can be chosen in one of n ways, the second in $n - 1$ ways, the third in $n - 2$ ways, and so on.

- recall the assignment for the caesar cipher:

plain: a b c d e f g h i j k l m n o p q r s t u v w x y z

cipher: d e f g h i j k l m n o p q r s t u v w x y z a b c

- if, instead, the “cipher” line can be any permutation of the 26 alphabetic characters, then there are $26!$ or greater than $4 * 10^{26}$ possible keys.
- this is 10 orders of magnitude greater than the key space for des and would seem to eliminate brute-force techniques for cryptanalysis.

(tabula recta)

KEY

PLAINTEXT

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

CIPHERTEXT

Poly Alphabetic	Mono Alphabetic
It is more secure in compare to mono-alphabetic.	It is less secure in compare to poly-alphabetic
More than one alphabets are used to substitution.	One single fixed alphabets are used substitution.
In this method, the substitution rule changes continuously from letter to letter according to the elements of the encryption key.	In this method, same substitution rule is use each substitution.
In this method, any one alphabets substitute with different alphabets using Vigenère table.	In this method, for a particular alphabet, only substitution can be used.

4. PLAY FAIR CIPHER:

- The **Playfair cipher** was the **first practical digraph substitution cipher**. The scheme was invented in **1854** by **Charles Wheatstone** but was named after Lord Playfair who promoted the use of the cipher.
- In playfair cipher unlike traditional cipher we encrypt a pair of alphabets(digraphs) instead of a single alphabet.

It was used for tactical purposes by British forces in the Second **Boer War** and in **World War I** and for the same purpose by the **Australians** during **World War II**.

Advantages:

- fast to use** and
- requires **no special equipment**.

FOR THE ENCRYPTION PROCESS LET US CONSIDER THE FOLLOWING EXAMPLE:

Key: monarchy

Plaintext: instruments

The Playfair Cipher Encryption Algorithm:

The Algorithm consists of 2 steps:

1. Generate the key Square(5×5):

1. The key square is a **5×5 grid of alphabets** that acts as the key for encrypting the plaintext. Each of the 25 alphabets must be **unique and one letter of the alphabet** (usually J) is omitted from the table (as the table can hold only 25 alphabets).
2. **If the plaintext contains J, then it is replaced by I.**
3. The initial alphabets in the key square are the **unique alphabets** of the key in the order in which they appear **followed by the remaining letters of the alphabet in order.**

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

2. Algorithm to encrypt the plain text:

the plaintext is split into **pairs of two letters** (digraphs). if there is **an odd number of letters**, a **z** is added to the last letter.

- FOR EXAMPLE:

PLAINTEXT: "INSTRUMENTS"

AFTER SPLIT: 'IN' 'ST' 'RU' 'ME' 'NT' 'SZ'

1. pair **cannot be made with same letter**. break the letter **in single** and add a **bogus letter** to the previous letter.

PLAIN TEXT: "HELLO"

AFTER SPLIT: 'HE' 'LX' 'LO'

- here **'x'** is the bogus letter.

- 2. if the letter is standing alone in the process of pairing, then add an extra bogus letter with the alone letter

PLAIN TEXT: "HELLOE"

AFTERSPLIT: 'HE' 'LX' 'LO' 'EZ'

- here 'z' is the bogus letter.

RULES FOR ENCRYPTION:

- if both the letters are in the same column: take the letter below each one (going back to the top if at the bottom).

- for example:

DIAGRAPH: "ME"

ENCRYPTED TEXT: CL

ENCRYPTION:

M -> C

E -> L

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

- if both the letters are **in the same row: take the letter to the right of each one** (going back to the leftmost if at the rightmost position).
- for example:

DIAGRAPH: "ST"

ENCRYPTED TEXT: TL

- ENCRYPTION:

S -> T

T -> L

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

- if neither of the above rules is true:
- form a rectangle with the two letters and take the letters on the horizontal opposite corner of the rectangle.
- for example:

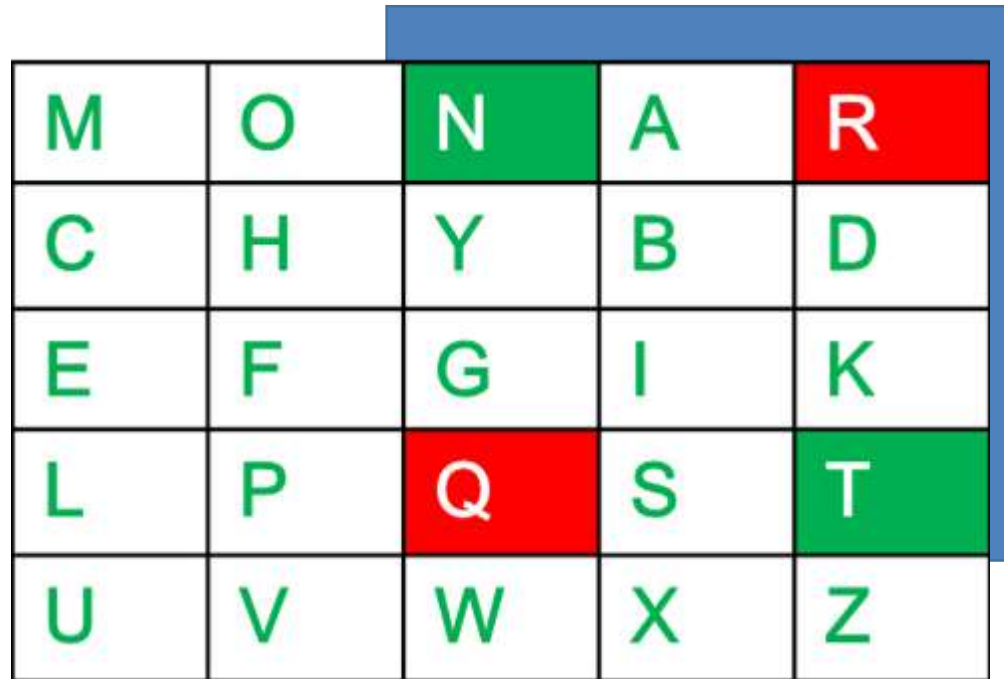
DIAGRAPH: "NT"

ENCRYPTED TEXT: RQ

- ENCRYPTION:

N -> R

T -> Q



M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

FOR EXAMPLE:

- PLAIN TEXT: "INSTRUMENTS"
- ENCRYPTED TEXT: GATLMZCLRQTX

in:	<table> <tr><td>M</td><td>O</td><td>N</td><td>A</td><td>R</td></tr> <tr><td>C</td><td>H</td><td>Y</td><td>B</td><td>D</td></tr> <tr><td>E</td><td>F</td><td>G</td><td>I</td><td>K</td></tr> <tr><td>L</td><td>P</td><td>Q</td><td>S</td><td>T</td></tr> <tr><td>U</td><td>V</td><td>W</td><td>X</td><td>Z</td></tr> </table>	M	O	N	A	R	C	H	Y	B	D	E	F	G	I	K	L	P	Q	S	T	U	V	W	X	Z	st:	<table> <tr><td>M</td><td>O</td><td>N</td><td>A</td><td>R</td></tr> <tr><td>C</td><td>H</td><td>Y</td><td>B</td><td>D</td></tr> <tr><td>E</td><td>F</td><td>G</td><td>I</td><td>K</td></tr> <tr><td>L</td><td>P</td><td>Q</td><td>S</td><td>T</td></tr> <tr><td>U</td><td>V</td><td>W</td><td>X</td><td>Z</td></tr> </table>	M	O	N	A	R	C	H	Y	B	D	E	F	G	I	K	L	P	Q	S	T	U	V	W	X	Z	ru:	<table> <tr><td>M</td><td>O</td><td>N</td><td>A</td><td>R</td></tr> <tr><td>C</td><td>H</td><td>Y</td><td>B</td><td>D</td></tr> <tr><td>E</td><td>F</td><td>G</td><td>I</td><td>K</td></tr> <tr><td>L</td><td>P</td><td>Q</td><td>S</td><td>T</td></tr> <tr><td>U</td><td>V</td><td>W</td><td>X</td><td>Z</td></tr> </table>	M	O	N	A	R	C	H	Y	B	D	E	F	G	I	K	L	P	Q	S	T	U	V	W	X	Z
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U	V	W	X	Z																																																																												

Encryption:

i -> g
n -> a
s -> t
t -> l
r -> m
u -> z
m -> c
e -> l
n -> r
t -> q
s -> t
z -> x

DECRYPTION IN PLAY FAIR CIPHER:

- **Generate the key square(5×5) at the receiver's end:**
- Each of the **25 alphabets must be unique and one letter** of the alphabet (usually J) is omitted from the table (as the table can hold only 25 alphabets). **If the plaintext contains J, then it is replaced by I.**
- **the key square are the unique alphabets of the key in the order in which they appear followed by the remaining letters**
- **Algorithm to decrypt the cipher text:** the **cipher text is split into pairs** of two letters (digraphs).

***Note:** the **ciphertext** always have **even** number of characters.*

EXAMPLE: **Key:** monarchy
ciphertext: gatlmzclrqtx

- CIPHERTEXT: "GATLMZCLRQTX"
- AFTER SPLIT: 'GA' 'TL' 'MZ' 'CL' 'RQ' 'TX'

1. Rules for Decryption:

- If both the letters are in the same column: Take the letter above each one (going back to the bottom if at the top).
For example:

Diagram: "cl"
Decrypted Text: me
Decryption:
c -> m
l -> e

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

-
- **If both the letters are in the same row:** Take the letter to the left of each one (going back to the rightmost if at the leftmost position).

For example:

Diagraph: "tl"

Decrypted Text: st

Decryption:

t -> s

l -> t

-

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

-
- **If neither of the above rules is true:** Form a rectangle with the two letters and take the letters on the horizontal opposite corner of the rectangle.
For example:

Diagraph: "rq"

Decrypted Text: nt

Decryption:

r -> n

q -> t

-

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

Plain Text: "gatlmzclrqtx"

Decrypted Text: instrumentsz

Decryption:

(red)-> (green)

ga -> in

tl -> st

mz -> ru

cl -> me

rq -> nt

tx -> sz

in:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

st:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

ru:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

me:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

nt:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

sz:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

5. Hill Cipher

- Hill cipher is a **poly alphabetic cipher** introduced by **Lester Hill in 1929**. Let us discuss the technique of hill cipher.
- **Plain text:** Binary
- **Key:** HILL
- **Choose the key** in such a way that it always **forms a square matrix**. With **HILL** as the key, we can form a **2×2 matrix**.
- Now, of plain text, you have to form **a column vector of length similar to the key matrix**.
- In our case, the key matrix is **2×2** then the column vectors of plain text would be **2×1**.

The general equation to find cipher text using hill cipher is as follow:

$$C = KP \text{ mod } 26$$

$$(c_1 \ c_2) = \begin{pmatrix} k_1 & k_2 \\ k_3 & k_4 \end{pmatrix} \begin{pmatrix} p_1 \\ p_2 \end{pmatrix} \text{ mod } 26$$

For our example, our key matrix would be:

$$\begin{pmatrix} H & I \\ L & L \end{pmatrix}$$

And our plain text matrices of 2×1 will be as follow:

$$\begin{pmatrix} B \\ I \end{pmatrix} \begin{pmatrix} N \\ A \end{pmatrix} \begin{pmatrix} R \\ Y \end{pmatrix}$$

Now, we have to convert the key matrix and plain text matrices into numeric matrices. For that number the alphabets such as A=0, B=1, C=2,, Z=25. So, considering the alphabet numbering:

Key matrix will be:

$$K = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}$$

Plain text matrices would be:

$$\begin{pmatrix} 1 \\ 8 \end{pmatrix} \begin{pmatrix} 13 \\ 0 \end{pmatrix} \begin{pmatrix} 17 \\ 24 \end{pmatrix}$$

In the first calculation, we would get two cipher alphabets for plain text alphabet 'B' & 'I'.

$$(c_1 \ c_2) = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 1 \\ 8 \end{pmatrix} \text{ mod } 26$$

$$(c_1 \ c_2) = \begin{pmatrix} 71 \\ 99 \end{pmatrix} \text{ mod } 26$$

$$(c_1 \ c_2) = \begin{pmatrix} 71 \\ 99 \end{pmatrix} \text{ mod } 26$$

$$(c_1 \ c_2) = \begin{pmatrix} 19 \\ 21 \end{pmatrix}$$

$$(c_1 \ c_2) = \begin{pmatrix} T \\ V \end{pmatrix}$$

So, the cipher alphabet for plain text alphabet 'B' & 'I' is 'T' & 'V'. Similarly, we have to calculate ciphertext for remaining plain text. And then accumulate them to form the ciphertext.

The calculated ciphertext for 'Binary' using hill cipher is 'TVNNZJ'.

OR.....

Hill Cipher

- **Encryption:**

- **1Key Matrix Generation:**

- **2Matrix Representation of Plaintext:**

- Convert the plaintext into a numerical equivalent, mapping each letter to its position in the alphabet (e.g., A=0, B=1, ..., Z=25).
- Arrange the numerical values into a column matrix of size $n \times 1$.

- **Matrix Multiplication:**

- Multiply the key matrix with the plaintext matrix.

- $C = K \times P$

- **Where:**

- C is the resulting matrix (cipher text),
- K is the key matrix,
- P is the plaintext matrix.

- The multiplication is performed modulo 26.

- **Conversion to Cipher text:**

Decryption:

- **Key Matrix Inversion:** $K \times K^{-1} = I$
- **Matrix Representation of Cipher text:**
- **Matrix Multiplication:**
 - Multiply the inverted key matrix with the ciphertext matrix.
- $P = K^{-1} \times C$
- Where:
 - P is the resulting matrix (plaintext),
 - K^{-1} is the inverted key matrix,
 - C is the cipher text matrix.
- The multiplication is performed modulo 26.
- **Conversion to Plaintext:**

6. One-Time Pad

- **Condition:**
- **key length** should be **as long as the plain text** to prevent the **repetition of key**.
- Along with that, the **key** should be **used only once**.
- **new key** for each **new message** and
- of the **same length** as a **new message**.

Plain text: Binary

Key: Cipher

Now again convert the plain text and key into the numeric form. For that number the alphabets such as A=0, B=1, C=2,, Z=25. So, our plain text and key in numeric form would be:

Plain text: 1 8 13 0 17 24

Key: 2 8 15 7 4 17

Now, you have to add the number of the plain text alphabet, to the number of its corresponding key alphabet. That means, for this example, we will add:

$$B+C = 1+2 = 3$$

$$I+I = 8+8 = 16$$

$$N+P = 13+15 = 28$$

$$A+H = 0+7 = 7$$

$$R+E = 17+4 = 21$$

$$Y+R = 24+17 = 41$$

The resultant ciphertext numbers we get are (3, 16, 28, 7, 21, 41)

If the addition of any plain text number and the key number is >26 , then subtract only that particular number from 26. We have the addition of two pair of plain text number and a key number, greater than 26, i.e. $N+P=28$ & $Y+R=41$.

Subtract them by 26.

$$N+P = 28 - 26 = 2$$

$$Y+R = 41 - 26 = 15$$

So, the final **ciphertext numbers** are (3, 16, 2, 7, 21, 15). Now convert this number to alphabets assuming A to be numbered 0 and B to be 1.....Z to 25.

Ciphertext: dqchvp.

In this way, we can convert plain text to cipher text using a one-time pad.

TRANSPOSITION TECHNIQUES

- performing permutations on the plain text, i.e, Changing each character of plain text for each round.
- rearranging the order of characters in a message to change its meaning.
- [Rail fence transposition](#)
- [Columnar transposition](#)
- [Improved columnar transposition](#)
- [Book cipher/running key cipher](#)

1. RAIL FENCE TRANSPOSITION

- **Step 1:** **plain text**: sequence of diagonals.
- **Step 2:** **cipher text**: the text is read as a sequence of rows.

To understand this in a better way, let us take an example:

D		F		N		T		E		A		T		A		L
	E		E		D		H		E		S		W		L	

The cipher text is then read off by writing the top row first, followed by the bottom row, to get "dfnteatal eedheswl".

ENCRYPTION

- Write plain Text **zigzag lines** across the page, and then read off each row.
- Firstly, you need to **have a key=number of rows** you are going to have.
- For the plaintext we used above, "**defend the east wall**", with a **key of 3**, we get the encryption process shown below.
- Note that at the end of the message we have inserted two "**x**"s.
- These are called **nulls, and act as placeholders**. We do this to make the message fit neatly in to the grid
- The cipher text is read off row by row to get "**dnetl eedheswlx ftaax**".

D				N				E				T				L		
	E		E		D		H		E		S		W		L		X	
		F				T				A				A				X

DECRYPTION

- We start writing the message, **but leaving a dash in place of the spaces yet to be occupied.**
- We start by making a grid with as many rows as the key is, and as many columns as the length of the cipher text.
- **We then place the first letter in the top left square, and dashes diagonally downwards where the letters will be. When we get back to the top row, we place the next letter in the ciphertext.**

Example-2

- For example, if you receive the ciphertext "**tekoohracirmnreatanftetytghh**", encrypted with a key of **4**, you start by placing the "t" in the first square. **You then dash the diagonal down spaces until you get back to the top row, and place the "E" here.** Continuing to fill the top row you get the pattern below.

T				E				K				O				O			
	-			-		-		-		-		-		-		-		-	
		-		-			-	-			-	-			-	-			-
			-				-				-				-				-

- **Example-3** CONTINUING THIS ROW-BY-ROW, WE GET THE SUCCESSIVE STAGES SHOWN BELOW.

TEKOOHRACIRMNREATANFTETGTGHH

T					E					K					O					O		
	H				R		A			C		I			R		M			N		R
		-		-			-		-			-		-			-		-			-
			-					-					-					-				-

The second stage in the decryption process.

T					E					K					O					O		
	H				R		A			C		I			R		M			N		R
		E		A			T		A			N		F			T		E			T
			-					-					-					-				-

The Third stage in the decryption process.

T					E					K					O					O		
	H				R		A			C		I			R		M			N		R
		E		A			T		A			N		F			T		E			T
			Y					T					G					H				H

The fourth and final stage in the decryption process.
Theyareattackingfromthenorth

Encryption steps

- **Key Selection:**
 - Choose the **number of rails (rows) for the transposition**. This determines how many rows you will use to write the message.
- **Message Arrangement:**
 - Write the message diagonally across the chosen number of rails.
- **Read Off Rows:**
 - Read the characters row by row to obtain the cipher text.

Decryption Steps:

- **Key Selection:**
 - Choose the same **number of rails** used for encryption.
- **Message Reconstruction:**
 - Write the ciphertext diagonally across the chosen number of rails.
- **Read Off Columns:**
 - Read the characters column by column to obtain the original message.

2. COLUMNAR TRANSPOSITION CIPHER

- Columnar transposition involves **writing the plaintext out in rows**, and then **reading the cipher text off in columns**.
- It is the **route cipher** where the route is to read down each column in order.
- For example, the plaintext "**a simple transposition**" with **5** columns looks like the grid below
- If we now read down each column we get the cipher text "ALNI SEST ITPI MROO PASN".

A	S	I	M	P
L	E	T	R	A
N	S	P	O	S
I	T	I	O	N

ENCRYPTION:

- We first pick a **keyword** for our encryption.
- We write the plaintext.
- Do numbering for columns acc. To alphabet order

T	O	M	A	T	O
5	3	2	1	6	4
T	H	E	T	O	M
A	T	O	I	S	A
P	L	A	N	T	I
N	T	H	E	N	I
G	H	T	S	H	A
D	E	F	A	M	I
L	Y	X	X	X	X

- **Encrypt the message:** "the tomato is a plant in the nightshade family" using the keyword **tomato**.
- check numbering for O and T.

- Starting with the column headed by "A", our ciphertext begins "TINESAX" from this column.
- We now move to the column headed by "M", and so on through the letters of the keyword in alphabetical order to get the cipher text **"TINESAX / EOAHTFX / HTLTHEY / MAIIAIX / TAPNGDL / OSTNHMX"** (where the / tells you where a new column starts). The final ciphertext is thus "TINES AXEOA HTFXH TLTHE YMAII AIXTA PNGDL OSTNH MX".

T	O	M	A	T	O
5	3	2	1	6	4
T	H	E	T	O	M
A	T	O	I	S	A
P	L	A	N	T	I
N	T	H	E	N	I
G	H	T	S	H	A
D	E	F	A	M	I
L	Y	X	X	X	X

DECRYPTION

- Firstly, if nulls have been used, then you start by writing out the keyword and the **alphabetical order of the letters of the keyword**.
- **You must then divide the length of the ciphertext by the length of the keyword.**
- The answer to this is the number of rows you need to add to the grid. **You then write the ciphertext down the first column until you reach the last row.** The next letter becomes the first letter in the second column (by the alphabetical order of the keyword), and so on.

- As an example, we shall decrypt the ciphertext
- "**aresa sxost heylo iiaie xpeng dlta htfax tenhm wx**" given the keyword *potato*.
- We start by writing out the keyword and the order of the letters.
- There **are 42** letters in the ciphertext, and the keyword has six letters, so we need **$42 \div 6 = 7$** rows.

[illegible]

ARESA SXOST HEYLO IIAIE XPENG DLLTA HTFAX TENHM WX

- Now we start by filling in the columns in the order given by the alphabetical order of the keyword, starting with the column headed by "A". After the first column is entered we have the grid shown to the right.
- We continue to add columns in the order specified by the keyword.

P	O	T	A	T	O
4	2	5	1	6	3
			A		
			R		
			E		
			S		
			A		
			S		
			X		

P	O	T	A	T	O
4	2	5	1	6	3
	O		A		
	S		R		
	T		E		
	H		S		
	E		A		
	Y		S		
	L		X		

P	O	T	A	T	O
4	2	5	1	6	3
	O		A		O
	S		R		I
	T		E		I
	H		S		A
	E		A		I
	Y		S		E
	L		X		X

P	O	T	A	T	O
4	2	5	1	6	3
P	O	T	A	T	O
E	S	A	R	E	I
N	T	H	E	N	I
G	H	T	S	H	A
D	E	F	A	M	I
L	Y	A	S	W	E
L	L	X	X	X	X

Now we read off the plaintext row at a time to get "potatoes are in the nightshade family as well".

- To decrypt the ciphertext "**ARESA SOSTH EYLOI IAIEP ENGD LTAHT FATEN HMW**", we start similarly to above, by heading the columns with the keyword *potato*.
- This time, to find how many rows we need, we do $38 \div 6 = 6.3333$. We round this up to the next number, which is 7, so we need 7 rows.
- When we multiply 6×7 we get 42, and $42 - 38 = 4$. Hence we need 4 placeholders in the last row. We get the grid below to the left. After plugging the ciphertext letters in, in the same way as above, we get the grid on the right.

[illegible]

P	O	T	A	T	O
4	2	5	1	6	3
P	O	T	A	T	O
E	S	A	R	E	I
N	T	H	E	N	I
G	H	T	S	H	A
D	E	F	A	M	I
L	Y	A	S	W	E
L	L				

Finally, we read off the plaintext in rows, to reveal the same plaintext as the other example, "potatoes are in the nightshade family as well".

IMPROVED COLUMNAR TRANSPOSITION

- it is similar to the basic columnar technique but is introduced with an improvement. the basic columnar technique is performed over the plain text but more than once. the steps for columnar technique with multiple rounds are as follow:

Step 1: The plain text is written in the rectangle of predetermined size row by row.

Step 2: To obtain the cipher text, read the plain text in the rectangle, column by column. Before reading the text in rectangle column by column, permute the order of columns the same as in basic columnar technique.

Step 3: To obtain the final cipher text repeat the steps above multiple time.

- let us discuss one example of a columnar transposition technique for better understanding. we will consider the same example of a basic columnar technique which will help in understanding the complexity of the method:

T	O	M	A	T	O
5	3	2	1	6	4
T	H	E	T	O	M
A	T	O	I	S	A
P	L	A	N	T	I
N	T	H	E	N	I
G	H	T	S	H	A
D	E	F	A	M	I
L	Y	X	X	X	X

T	O	M	A	T	O
5	3	2	1	6	4
T	I	N	E	S	A
X	E	O	A	H	T
F	X	H	T	L	T
H	E	Y	M	A	I
I	A	I	X	T	A
P	N	G	D	L	O
S	T	N	H	M	X

We put the plaintext into the grid below the keyword tomato to get the cipher text "**TINES AXEOA HTFXH TLTHE YMAII AIXTA PNGDL OSTNH MX**"

We now write the cipher text retrieved from the grid to the left in the grid in rows as if it was the plaintext.

After this **double transposition**, we get the cipher text "**EATMX DHNOH YIGNI EXEAN TATTI AOXTX FHIPS SHLAT LM**".

This double transposition increases the security of the cipher significantly. It could also be implemented with a different keyword for the second iteration of the cipher. In fact, until the invention of the VIC Cipher, Double Transposition was seen as the most secure cipher for a field agent to use reliably under difficult circumstances.

Book Cipher or Running Key Cipher

- The book cipher or the running key cipher works on the basic principle of one-time pad cipher. In onetime pad cipher the key is taken as long as the plain text and is discarded after the use. Every time a new key is taken for a new message.
- The improvement to the onetime pad in Book cipher is that the key or the onetime pad is taken from the book. Let us discuss the steps:
- **Step 1:** Convert the plain text in numeric form consider A=0, B=1, C=3 ..., Z=25.
- **Step 2:** Take an onetime pad or key from any of the books and convert it in the numeric form also. But the key must be as long as the length of plain text.
- **Step 3:** Now add the numeric form of both plain text and key, each plain text letter with corresponding key text letter. If the addition of any plain text letter with corresponding key text letter is >26 , then subtract it with 26.
- Let us understand with the example:
- **Plain text:** Meet Tomorrow
- **Key** taken from the book: ANENCRYPTION.
- Now we have to convert this plain text and key text in numeric form and add them to get cipher text as shown in the image below:

Numeric form
Plain Text

M E E T T O M O R R O W
12 4 4 19 19 14 12 14 17 17 14 22

Numeric form
Key Text

A N E N C R Y P T I O N
0 13 4 13 2 17 24 15 19 8 14 13

Add the numeric form of
Plain text and Key Text:

+ 12 4 4 19 19 14 12 14 17 17 14 22
0 13 4 13 2 17 24 15 19 8 14 13

Subtract Numbers
> 26 by 26

12 17 8 32 21 31 36 29 36 25 28 35

12 17 8 6 21 5 10 3 10 25 3 9

New Cipher
Text

M R I G V F K D K Z D J

The **cipher text** obtained is MRIGVFKDKZDJ.

Stenography application and its limitation

- It is the act of recording spoken words through shorthand using a stenotype machine.
- Stenography is the process of writing in shorthand or a system of shorthand writing.
- a shorthand system:
 - symbols->specific sounds, concepts, or letters
 - Letters->meanings.
- goal is to record the spoken word by copying it.
- Applications:
 - court reporters to record proceedings and events much quicker than they would be able to do using a standard keyboard.
 - less laborious and more accurate manner.
 - Explore Other Education Options Related to Criminal Justice and Legal Studies
 - Working as a Stenographer.

Limitations:

- capacity
- It is challenging to develop an efficient **shorthand system**.
- that can **handle large amounts** of data
- while maintaining security and **hiding information** effectively
- *Stenography*: the action or process of writing in shorthand or taking dictation.
- *Steganography*: the practice of concealing messages or information within other non-secret text or data.

WHAT IS STEGANOGRAPHY?

- A steganography technique involves **hiding sensitive information within an ordinary, non-secret file or message**, so that it will not be detected.
- the sensitive information will then be **extracted from the ordinary file** or message at its destination, thus avoiding detection.
- steganography is an additional step that can be used in **conjunction with encryption** in order to conceal or protect data.
- use: to hide text, video, images, or even audio data.

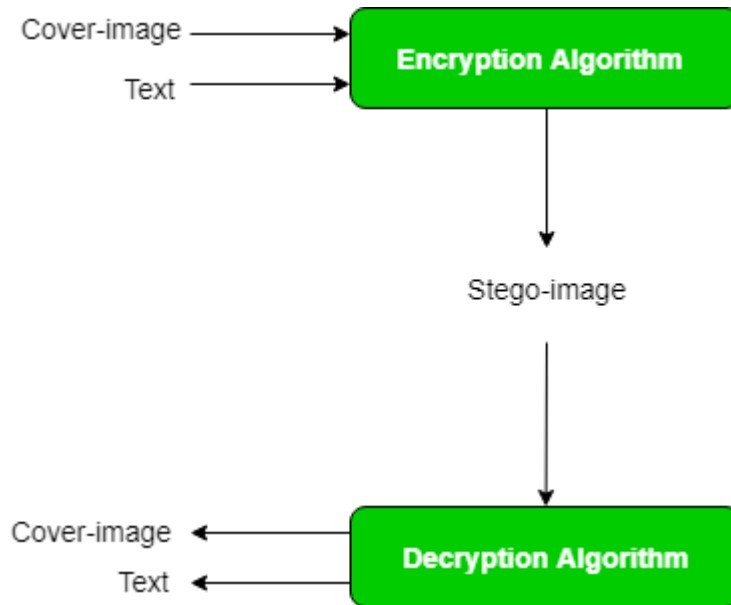


Figure 8: Normal image (was also used later to contain hidden data)



Figure 9: After StirMark filtering the image is still there but degraded heavily



ed off as Figure 8 with Steganographic content, after same levels as Figure 9 the image is destroyed.

DIFFERENT TYPES OF STEGANOGRAPHY

- 1. **text steganography**
- 2. **image steganography** –
- 3. **audio steganography** – it is the science of hiding data in sound. used digitally, it protects against unauthorized reproduction. **watermarking** is a technique that encrypts one piece of data (the message) within another (the "carrier"
- 4. **video steganography** – video steganography is a method of secretly embedding data or other files within a video file on a computer.
- 5. **network or protocol steganography** – it involves concealing data by using a network protocol like tcp, udp, icmp, ip, etc.,

STEGANOGRAPHY EXAMPLES INCLUDE

- writing with invisible ink
- embedding text in a picture (like an artist hiding their initials in a painting they've done)
- concealing information in either metadata or within a file header
- hiding an image in a video, viewable only if the video is played at a particular frame rate.

STEGANOGRAPHY APPLICATIONS

- image steganography: this application is a javascript tool used to hide images in other image files
- **openstego**: this program is an open-source steganography tool
- **xiao steganography**: xiao hides secret files
- **crypture**: this application is a command-line tool used to conduct steganography
- **noclue**: this application is an open-source tool that hides text information in both video and image carrier files
- **steganography master**: this app is an android-based open-source tool that can hide text in an image and gives you a decoding tool to pull hidden text messages from image files. it supports multiple image formats (bmp, jpg, ico, png)
- **steghide**: steghide is an application that hides data in different audio and image files, including jpeg

STEGANOGRAPHY VS. CRYPTOGRAPHY

Factors	Steganography	Cryptography
Explanation	It's a method to conceal the fact that communication is taking place	It's a method for making information unintelligible
Aim	Maintain communication security	Enable data protection
Key	Optional, but increases security when utilized	Necessary prerequisite
Data Visibility	No	Yes
Failure	Once hidden information is decoded, the data can be used by anyone	You can recover the original message from the ciphertext if you can access the decryption key

BLOCK CIPHERS AND MODES/METHODS OF OPERATIONS

- block cipher is an encryption algorithm that takes a **fixed size of input say b bits and produces a ciphertext of b bits again.**
- Input $> b$ bits it can be divided further.
- **ELECTRONIC CODE BOOK (ECB)**
- **CIPHER BLOCK CHAINING (CBC)**
- **CIPHER FEEDBACK MODE (CFB)**
- **OUTPUT FEEDBACK MODE (OFM)**
- **COUNTER MODE (CM)**

ELECTRONIC CODE BOOK (ECB) –

-

Electronic Code Book is simple mode.

- direct encryption of each block of input plaintext and output is in form of blocks of encrypted cipher text.
- Generally, if a message is larger than b bits in size, it can be broken down into a bunch of blocks and the procedure is repeated.

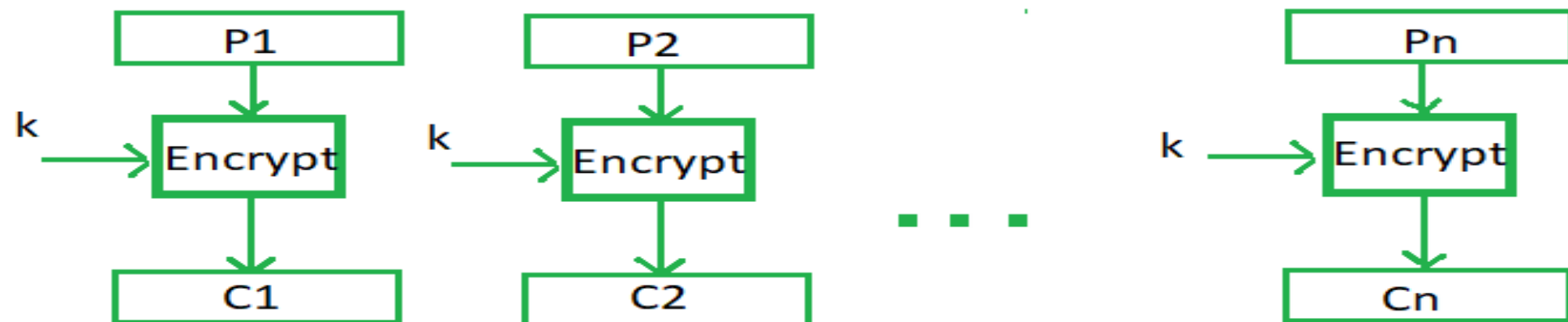
Advantages of using ECB –

- Parallel encryption of blocks of bits is possible, so it is faster.
- Simple way of the block cipher.

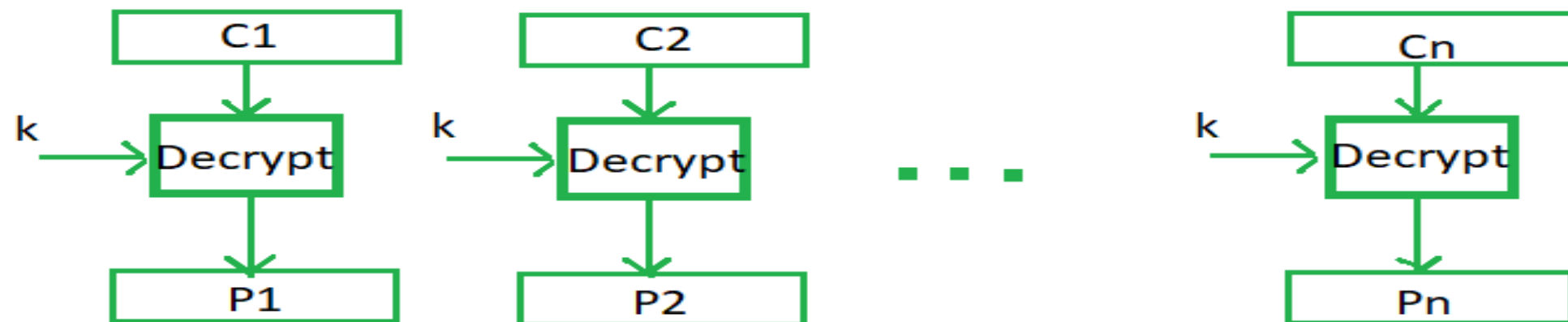
Disadvantages of using ECB –

- Prone to cryptanalysis since there is a direct relationship between plaintext and ciphertext.

Encryption



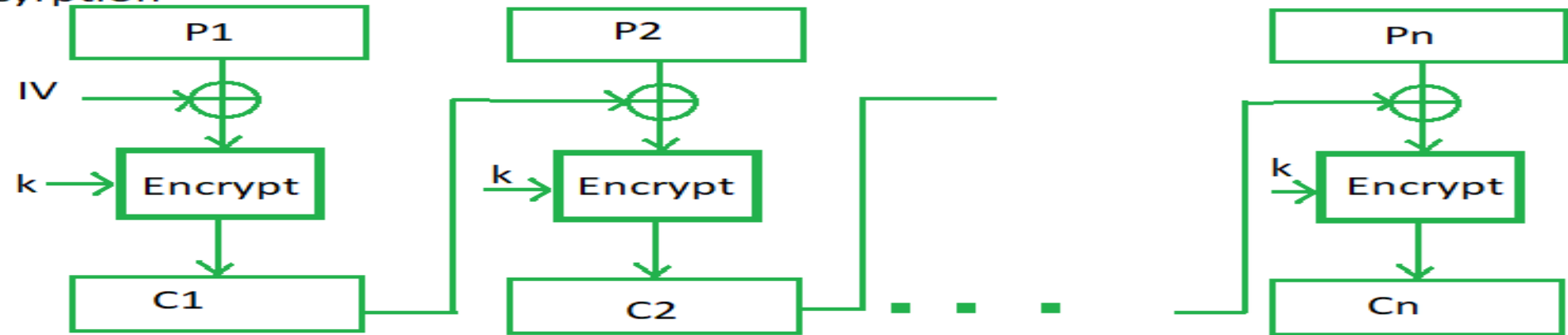
Decryption



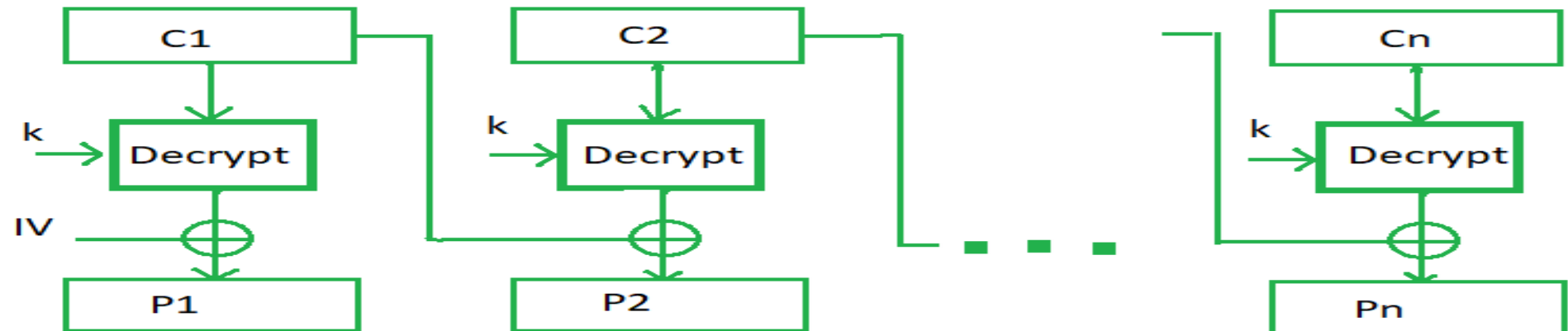
CIPHER BLOCK CHAINING –

- previous cipher block is given as input to the next encryption algorithm after xor with the original plaintext block. in a nutshell here, a cipher block is produced by encrypting an xor output of the previous cipher block and present plaintext block.
- advantages of cbc –
 - cbc works well for input greater than b bits.
 - cbc is a good authentication mechanism.
 - Protect against cryptanalysis.
- disadvantages of cbc –
 - parallel encryption is not possible since every encryption requires a previous cipher.

Encryption



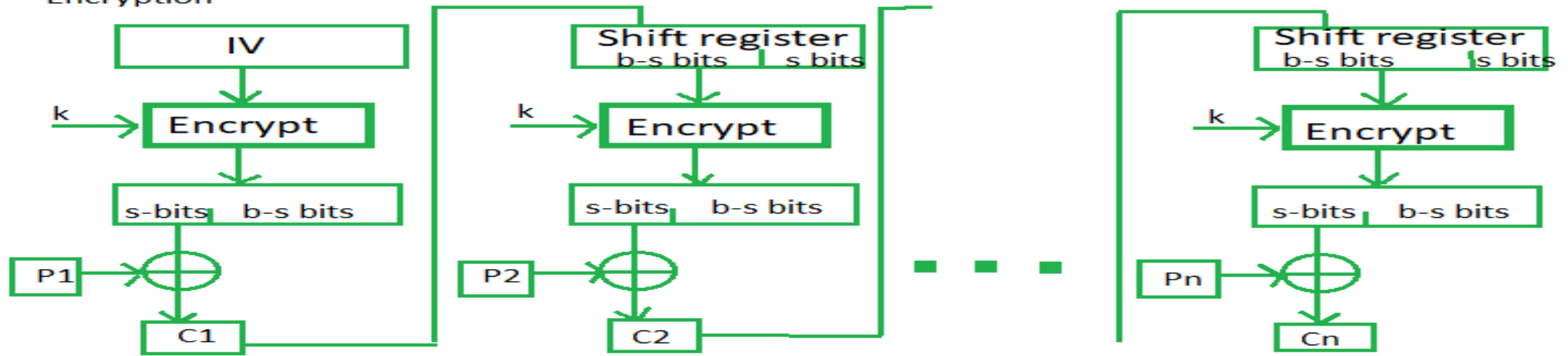
Decryption



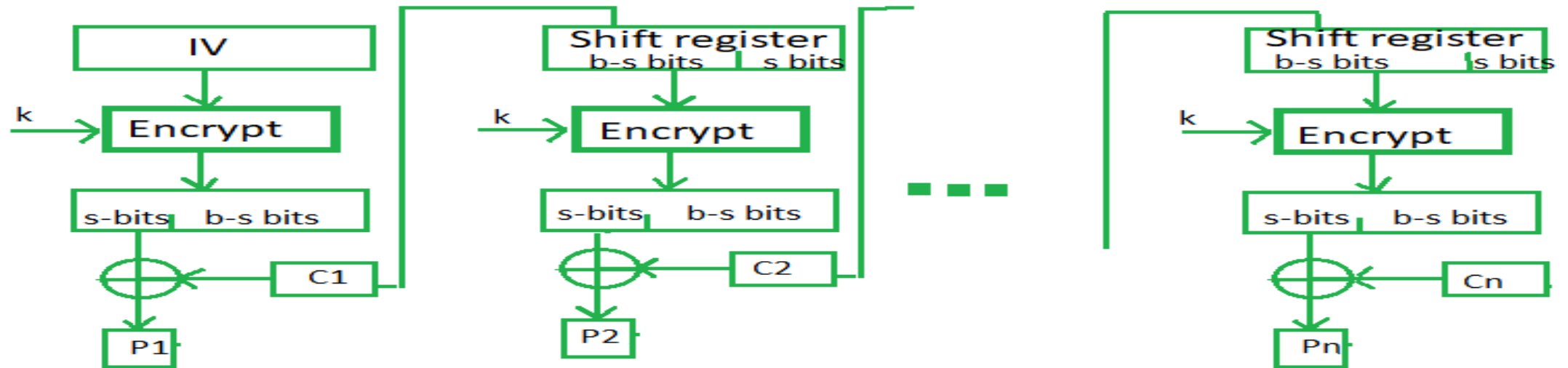
CIPHER FEEDBACK MODE (CFB) –

- cipher is given as **feedback to the next block** of encryption with some new specifications:
- first, an initial vector **iv** is used for first encryption and output bits are divided as a **set of s and $b-s$ bits**. The left-hand side s + plaintext bits to which an **xor** operation is applied.
- the result is given as **input to a shift register having $b-s$ bits to lhs, s bits to rhs** and the process continues.
- the encryption and decryption process for the same is shown below, both of them use encryption algorithms.
- **advantages of cfb** – **difficult** for applying cryptanalysis.
- **disadvantages of using cfb** – Parallelism not possible

Encryption



Decryption



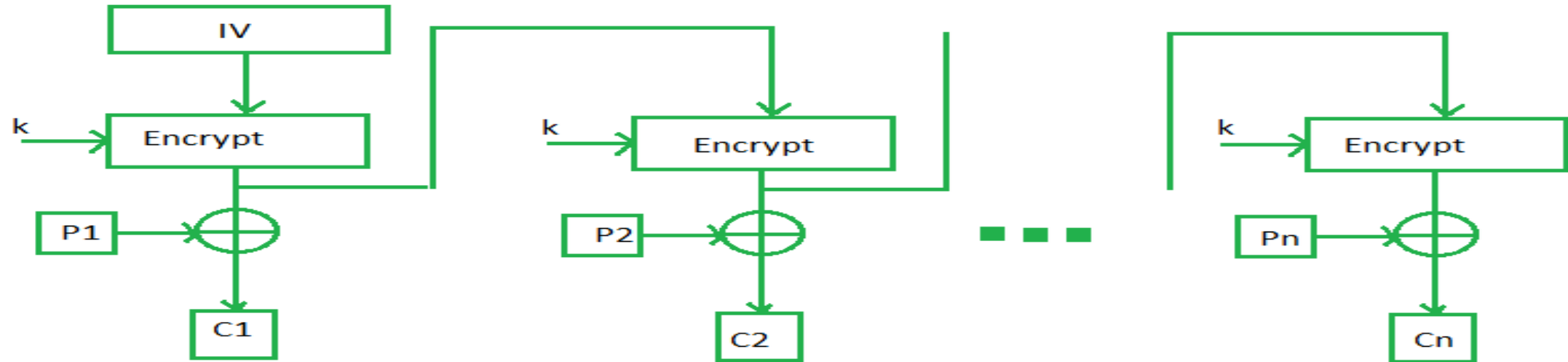
OUTPUT FEEDBACK MODE –

- the output feedback mode follows nearly the **same process as the cipher feedback mode** except that it sends the **encrypted output as feedback instead of the actual cipher which is xor output.**
- in this output feedback mode, all bits of the block are sent instead of **sending selected s bits**

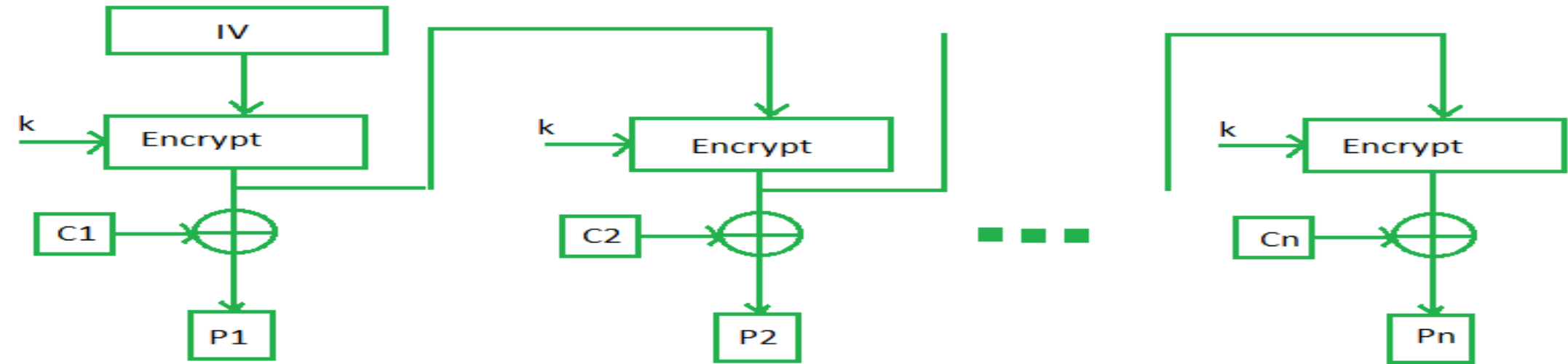
advantages of ofb –

- a single bit error in a block is propagated to all subsequent blocks. this problem is solved by ofb as it is free **from bit errors** in the plaintext block.
- **disadvantages of ofb-**
 - the drawback of ofb is that, because to its operational modes, it is more susceptible to a message stream modification attack than cfb.

Encryption



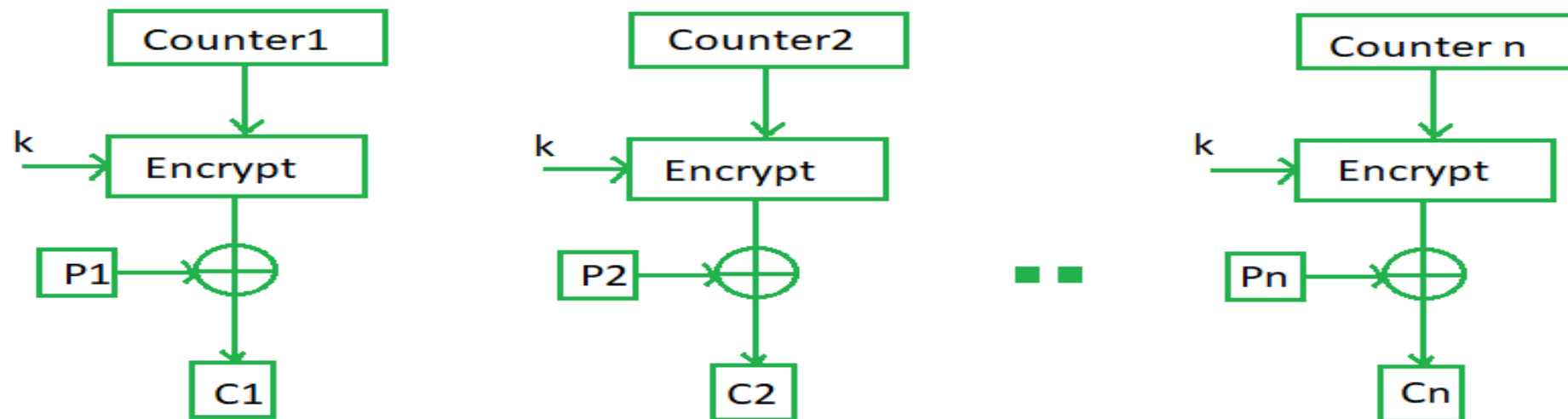
Decryption



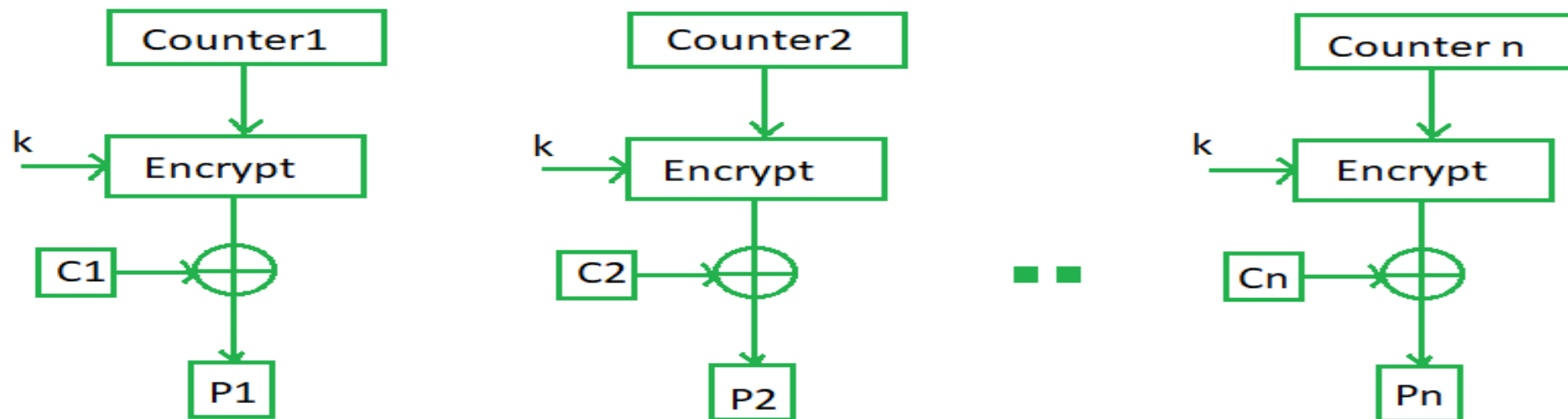
COUNTER MODE –

- Every time a counter-initiated value is encrypted and given as **input to xor with plaintext** which results in ciphertext block. the ctr mode is independent of feedback use and thus can be implemented in **parallel**
- **advantages**
 - since there is a different counter value for each block, **the direct plaintext and ciphertext relationship is avoided.**
 - **parallel execution of encryption is possible.**
- **disadvantages of counter-**
 - the fact that ctr mode requires a **synchronous counter** at both the transmitter and the receiver is a severe drawback.

Encryption



Decryption



- **data encryption:**

APPLICATIONS OF BLOCK CIPHERS

- **file and disk encryption:** block ciphers are used for encryption of entire files and disks in order to protect their contents and restrict from unauthorized users. the disk encryption softwares such as bitlocker, truecrypt aslo uses block cipher to encrypt data and make it secure.
- **virtual private networks (vpn):**
- **secure sockets layer (ssl) and transport layer security (tls):** ssl and tls protocols use block ciphers for encryption of data that is transmitted between **web browsers and servers over the internet..**
- **digital signatures:** block ciphers are used in the digital signature algorithms, to provide authenticity and integrity to the digital documents.
- this encryption process generates the unique signature for each document that is used for verifying the authenticity and detecting if any malicious activity is detected.
-

Feistel cipher

Feistel Cipher model is a **structure or a design used** to develop **many block ciphers** such as **DES**.

Feistel cipher may have **invertible, non-invertible and self invertible** components in its design.
Same encryption as well as decryption algorithm is used.

A **separate key is used** for each round. However **same round keys** are used for encryption as well as decryption.

Feistel cipher algorithm

- Create a list of all the **Plain Text** characters.
- Convert the **Plain Text to Ascii** and **then 8-bit binary format**.
- Divide the binary Plain Text string into two halves: left half (**L1**) and **right half (R1)**
- Generate a random binary keys (**K1 and K2**) of **length** equal to the half the length of the Plain Text for the two rounds.

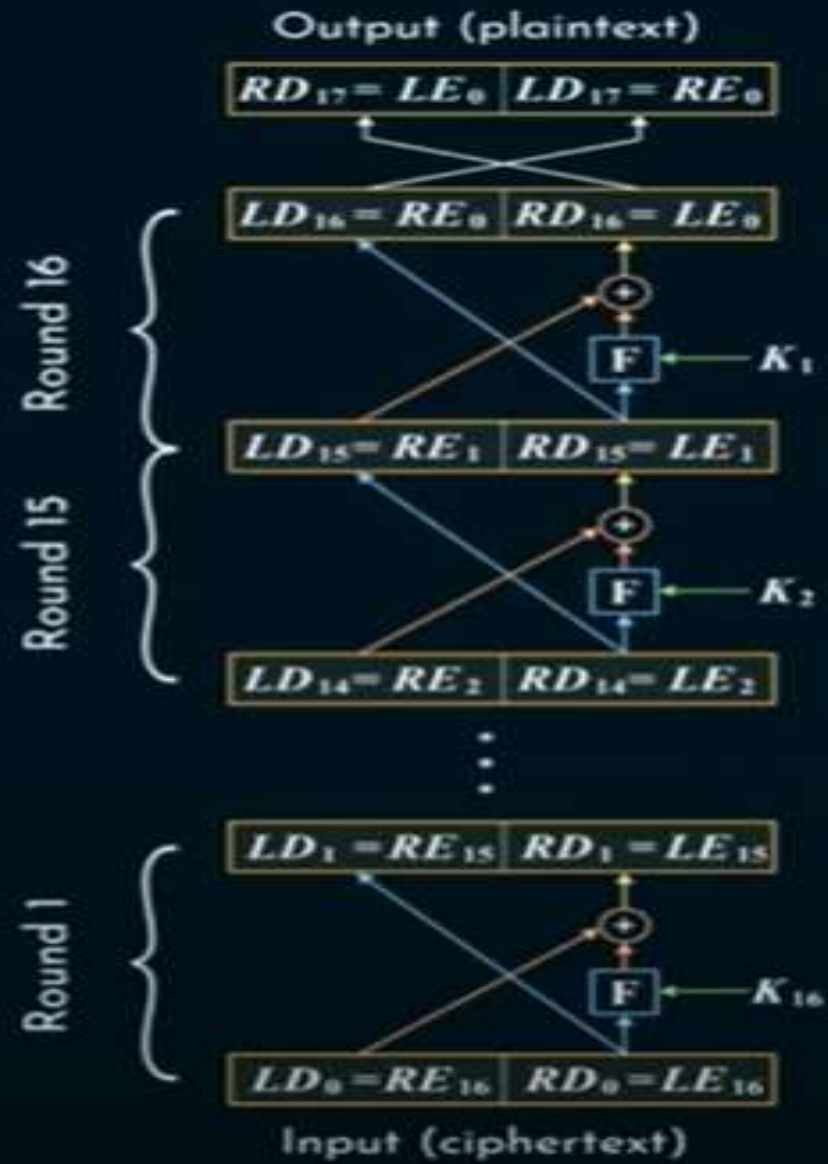
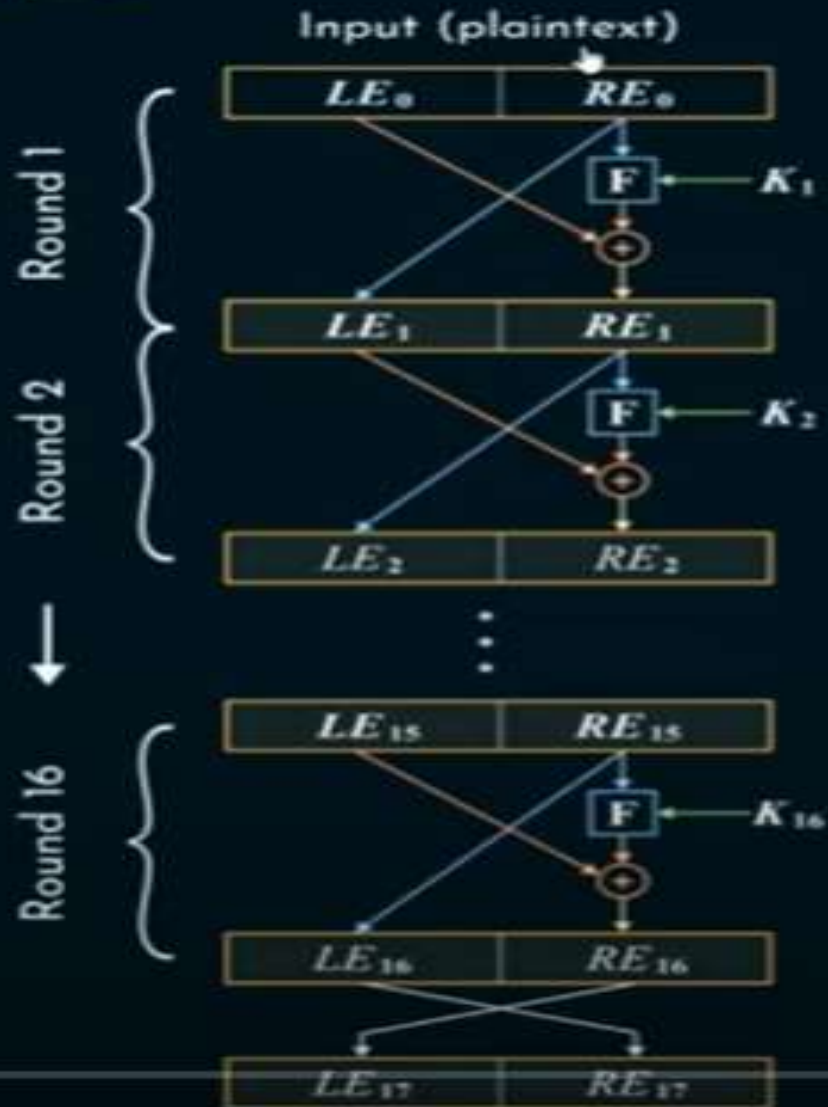
First Round

- **a.** Generate $f1 = \text{xor}(R1, K1)$
- **b.** $R2 = \text{xor}(f1, L1)$ $L2 = R1$

Second Round

- **a.** Generate function $f2 = \text{xor}(R2, K2)$
- **b.** $R3 = \text{xor}(f2, L2)$ $L3 = R2$
- Concatenation of $R3$ to $L3$ is the Cipher Text
- Same algorithm is used for decryption to retrieve the Plain Text from the Cipher Text.

Figure



Plain Text is: Hello

Cipher Text: E1!w(

Retrieved Plain Text is: 'Hello'

Plain Text is: Geeks

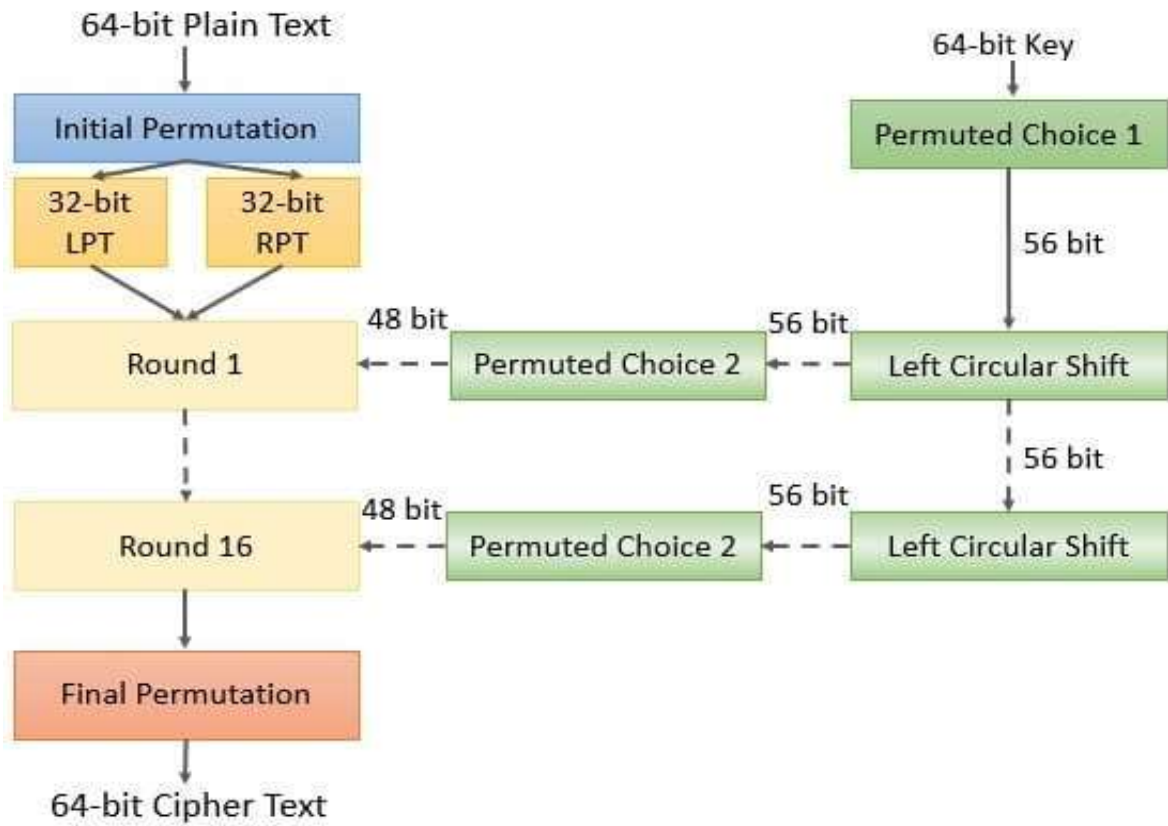
Cipher Text: O;Q

Retrieved Plain Text is: 'Geeks'

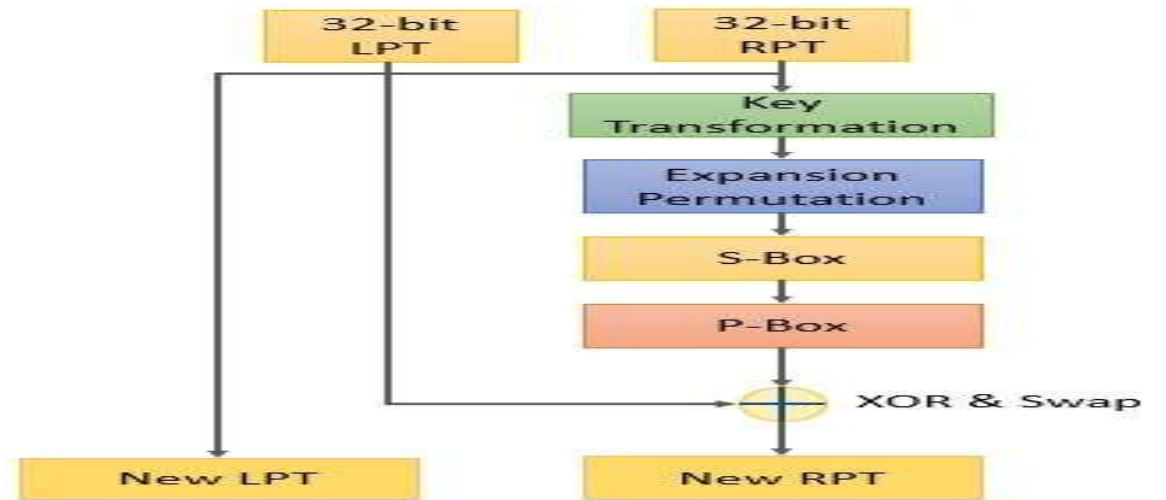
- Feistel cipher design divides blocks in left and right halves, right half with a sub-key and swaps the two halves.
- DES consists of 16 rounds using a 56-bit key from which 48- bit subkeys are generated.

Data Encryption Standard (DES)

- DES is a **symmetric block cipher**.
- DES encrypts **64-bit plain** text to **64-bit cipher** text.
- DES uses a 56-bit key for encryption.
- Encryption and decryption algorithm is the same in DES. But, the procedure of encryption is reversed while decryption.
- **16 rounds** in DES strengthens the algorithm.
- Each round has the same function which involves **key transformation, expansion permutation, s-box substitution, p-box permutation and XOR and swapping**.
- So, this is all about the DES block cipher. In 2001 the NIST has published **Advanced Encryption Standard** (AES) to replace DES. Because DES was meant for the hardware and its relatively **slow performance on software** leads to its replacement.



Data Encryption Standard (DES)



- **Encryption**
- **Step 1:** In the first step the 64-bit plain text undergoes **initial permutation** which rearranges the bits to produce **two 32-bit permuted block** which is called left plain text (**LPT 32-bit**) and right plain text (**RPT 32-bit**).
- **Step 2:** Now, **16 rounds** of DES encryption will be performed on this LPT and RPT with a **56-bit key**.
- **Step 3:** After the 16_{th} round the 32-bit LPT and 32-bit RPT are integrated which forms **a 64-bit block again and then the final permutation is applied to this 64-bit block, to obtain the 64-bit ciphertext.**

- **Decryption**
- The same Data Encryption Standard algorithm used for encrypting the plain text is also used to decrypting the ciphertext.
- But the algorithm is reversed, such as the initial and final permutation events are reversed. Even the sequence of the subkeys applied in 16 rounds of DES are also reversed.
- Rounds in DES
- Each round of DES performs the same function. So, below are the steps of the function performed in each round of DES algorithm:
- Key Transformation
- Expansion Permutation
- S-box Substitution
- P-box Permutation
- XOR and Swap

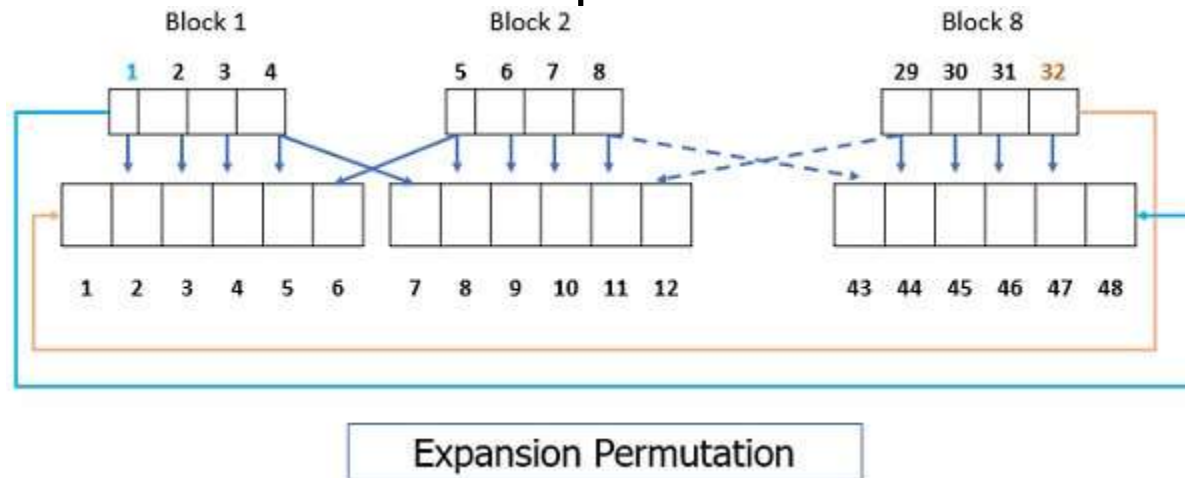
- **1. Key Transformation**

- Initial key size is 64-bit which is reduced to the 56-bit key. This is done by discarding every 8_{th} bit from the 64-bit key. So, for each round of DES, this 56-bit key is used.
- In the key transformation step, this 56-bit is transformed to the 48-bit key.
- Let's see how?
- The 56-bit key is split into two halves. Each half is of 28-bit. Now, a left circular shift is implemented on both the halves which shift the bits in two halves by one or two positions depending on the order of the round.
- If the round number is 1, 2, 9, 16 then the left circular shift is done by one position else, if any other order number is encountered the circular shift is done by two positions.

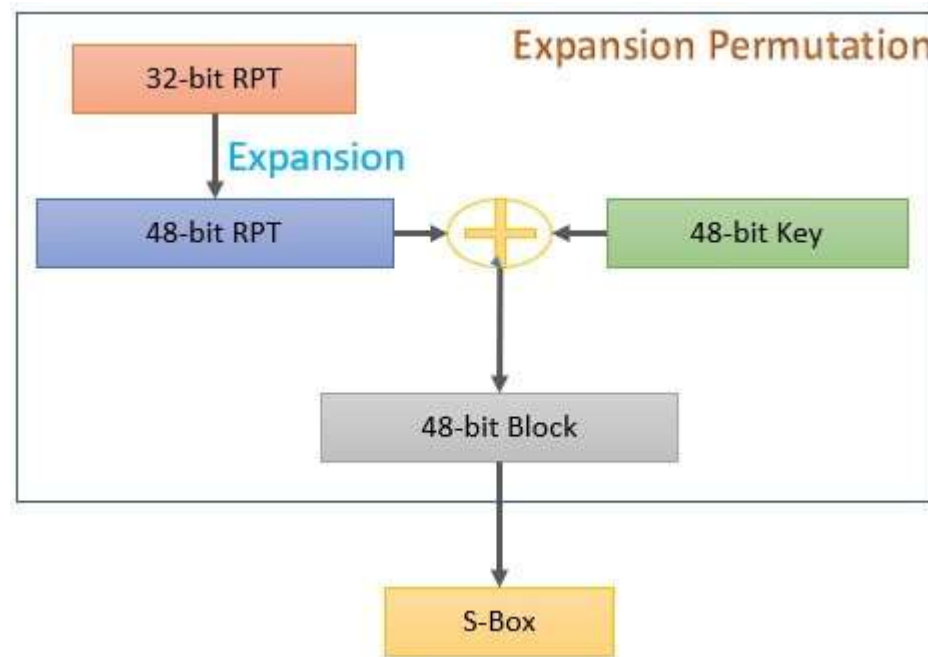
- After the left circular shift both the 28-bit halves are integrated and **permutation** is performed on a 56-bit key.
- And from this permuted 56-bit key **48-bit subkey** is selected.
- As the key transformation step involves permutation along with compression of the 56-bit key to 48-bit subkey it is also called **compression permutation**.
- Thus, the key transformation step provides a different 48-bit subkey for each round of DES.

- **2. Expansion Permutation**

- In the first step of encryption, during the initial permutation of DES, the 64-bit plain text is permuted and we have 32-bit LPT and 32-bit RPT.
- Now, the **expansion permutation** is performed on the **32-bit RPT** which transforms it from **32-bit to 48-bit**. The **32-bit LPT** is untouched during the process.
- For expansion, the 32-bit RPT is divided **8 blocks**, where each block is of **4-bit**. Each block of 4-bit is expanded to block of **6-bit** as you can see in the image below. The extreme two bits of the block are repeated and the middle two-bits is passed on as it is.

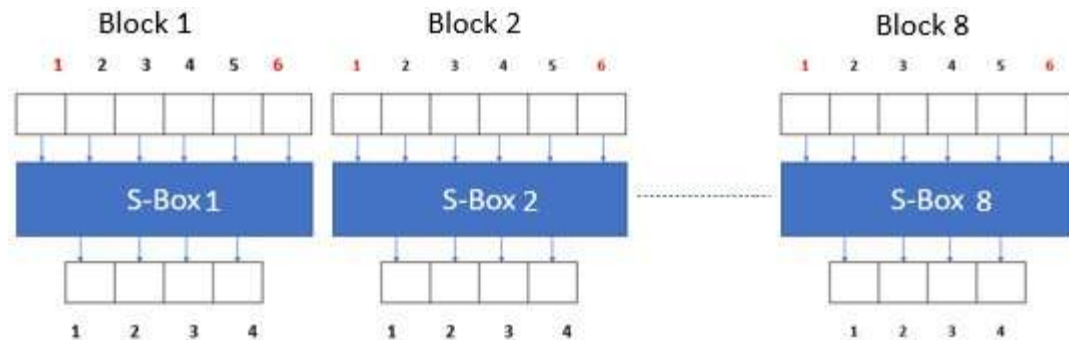


- After expansion, now the **48-bit RPT is XORed with the 48-bit key** and the resultant 48-bit block is transferred to S-box substitution.

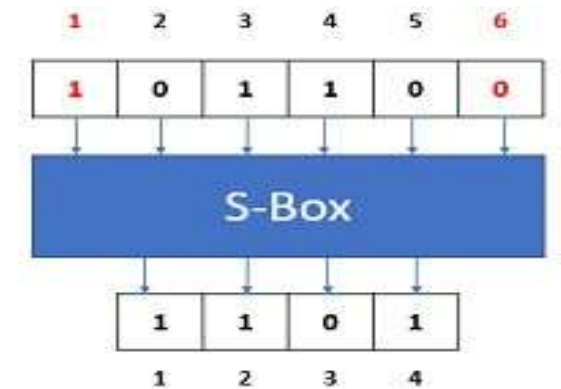


- **3. S-box Substitution**

- The input to S-box is **48-bit** resultant block of expansion permutation. In S-box substitution, the input **48-bit block is transformed to 32-bit** block
- but the question is how?
- The **48-bit input block** to S-box substitution is divided into **8 blocks of 6-bit each** and is provided as input to 8 substitution boxes also called **S-boxes**. Each s-box has an input of 6-bit and its **outputs 4-bit**.



- Each S-box is considered as the table which has **4 rows** 0-3 and **16 columns** i.e. 0-15. Why 4 rows and 16 columns? Because from the **6-bit input to s-box** the **two extreme bits** are used to determine the number of **rows**.
- 2 bits can form 4 combinations of 0s and 1s:
- $2^2 = 4$ rows
- The middle 4 bits are used to determine the number of columns
- $4^2 = 16$ columns
- Now, we will see how the **6-bit input is transformed to the 4-bit output**.
- For example, in the figure above you can see that input bit, at position 1 and 6 i.e. '10' that represent 2 which determines **2nd row** and middle 4 bits i.e. '0110' that represent 6 which determines the **6th column** of s-box. The value at the intersection of the 2nd row and 6th column will represent the 4-bit output of corresponding s-box.
-



14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
4	1	14	8	2	6	13	11	15	12	9	7	3	10	5	0
15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

S-Box Table 1

In our example, the intersection of the 2nd row and 6th column is 13 its **4bit representation** is **1101**. So, the output of S-box 1 is 1101.



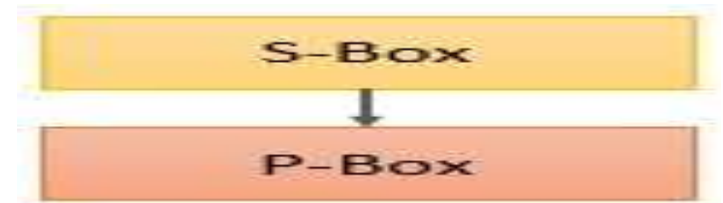
Similarly, the remaining S-boxes function will be carried out to find the remaining 4-bit outputs. In this way, the 8 s-boxes would generate 8, 4-bit output blocks which when integrated forms 32-bit output.

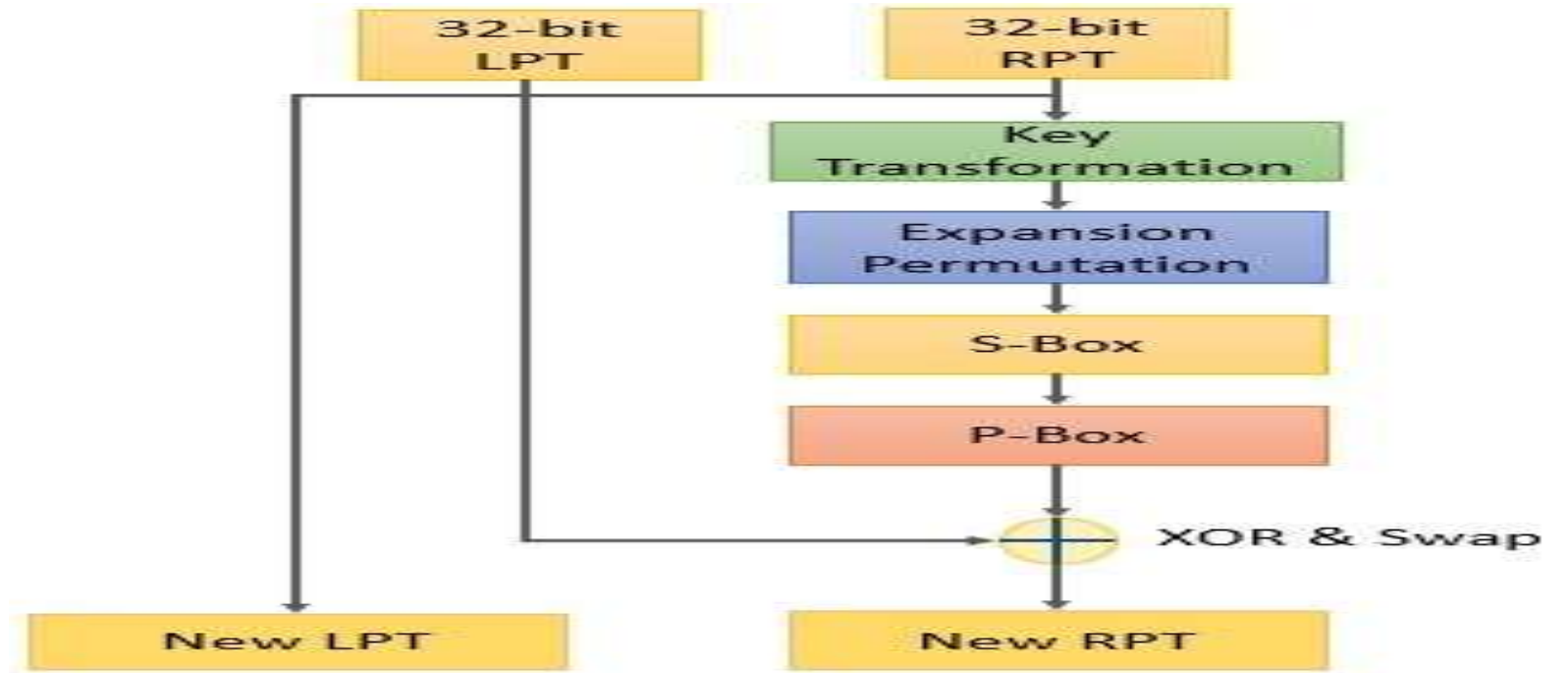
- **4. P-box Permutation**

- The 32-bit output obtained from s-box substitution is provided as an input to P-box. Here, the **32-bit input is simply permuted and send to the next step.**

- **5. XOR and Swap**

- As you can observe so far, we have been operating on the **32-bit RPT** of the initial 64-bit plain text. The 32-bit LPT is still untouched.
- In this step, the **32-bit LPT** of the initial **64-bit plain text** is **XOR** with the output of P-box permutation.
- The result of the XOR is the new RPT for next round and the old RPT is swapped with LPT. So, the old RPT is now new LPT for next round. o, the old RPT is now new LPT for next round.





- So, here the process of a single round ends and next round start which has the same function.
- After 16 such rounds, the output of the 16th round undergoes **final permutation**. The result of this final permutation is 64-bit ciphertext.

- **Advantages**

- DES has a 56-bit key which raises the possibility of 2^{56} possible keys which make **brute force impossible**.
- it impossible for any to discover the design of the s-boxes which makes the attack more impossible.
- The number of rounds in DES increases the **complexity of the algorithm**.

- **Disadvantages of DES**

- The DES algorithm is less efficient **when implemented in software**, resulting in slower performance.
- However, the **cryptanalysis attack** is easier than the brute force attack on DES.

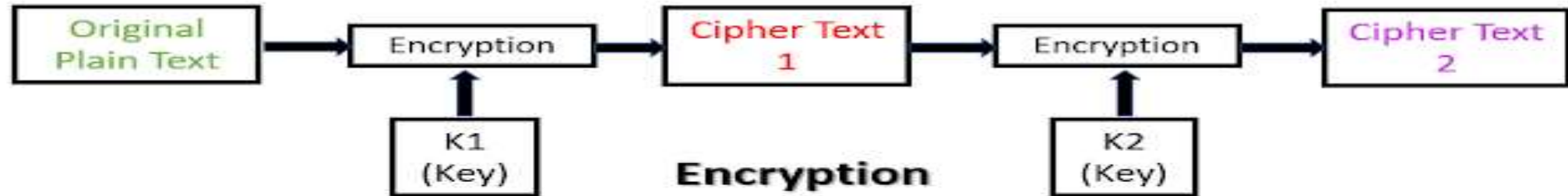
DOUBLE DES AND TRIPLE DES

DOUBLE DES

- ❑ In this approach, we use two instances of DES ciphers for encryption and two instances of reverse ciphers for decryption.
- ❑ Each instances use a different key.
 - The size of the key is doubled.
- ❑ There are issues of reduction to single stage.
- ❑ However, double DES is vulnerable to meet-in-the-middle attack.

Double DES Encryption

- Double DES performs the same operations as DES only difference is that double DES use two keys K1 & K2.



Mathematically double DES encryption is represented as,

1. $C1 = E(K1, P)$
 2. $C2 = E(K2, C1)$
- $$C2 = E(K2, E(K1, P))$$

Double DES Decryption

- Decryption of double DES is **reverse of encryption**.
- First it performs decryption on cipher text C2, which is decrypted using K2 and obtains cipher text C1.
- Again, cipher text C1 is decrypted by using another key K1 & generate original plain text P.



Mathematically double DES decryption is represented as,

1. $C1 = D(K2, C2)$

2. $P = D(K1, C1)$

$P = D(K1, D(K2, C2))$

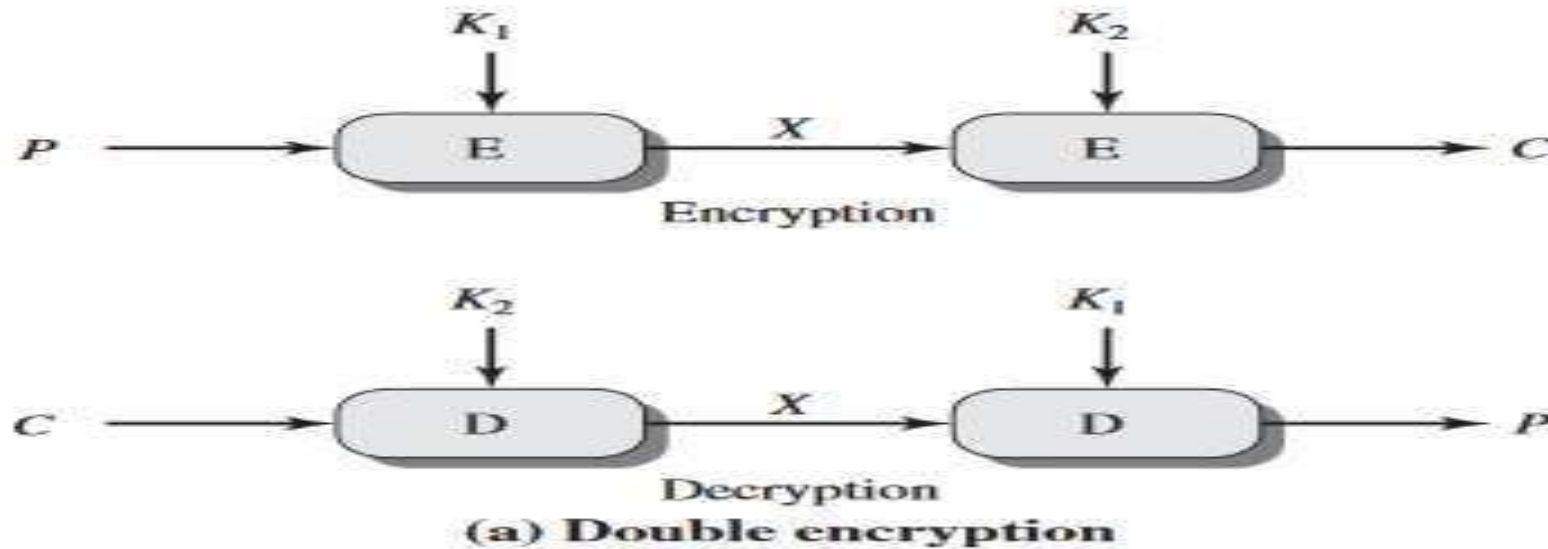
DOUBLE DES

- ? Given a plaintext P and two encryption keys K_1 and K_2 , a cipher text can be generated as,

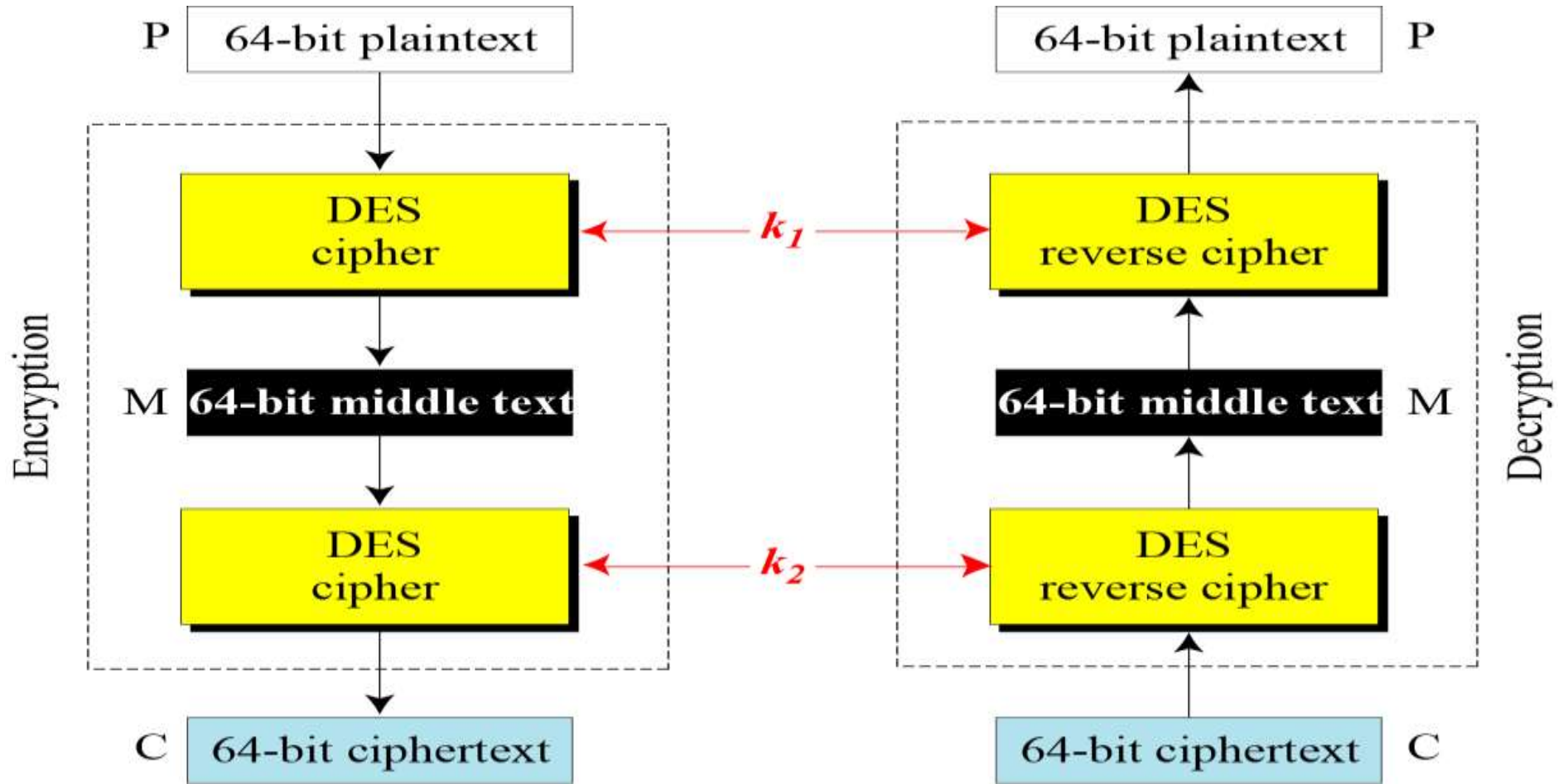
$$C = E(K_2, E(K_1, P))$$

- ? Decryption requires that the keys be applied in reverse order,

$$P = D(K_1, D(K_2, C))$$



MEET-IN-THE-MIDDLE ATTACK



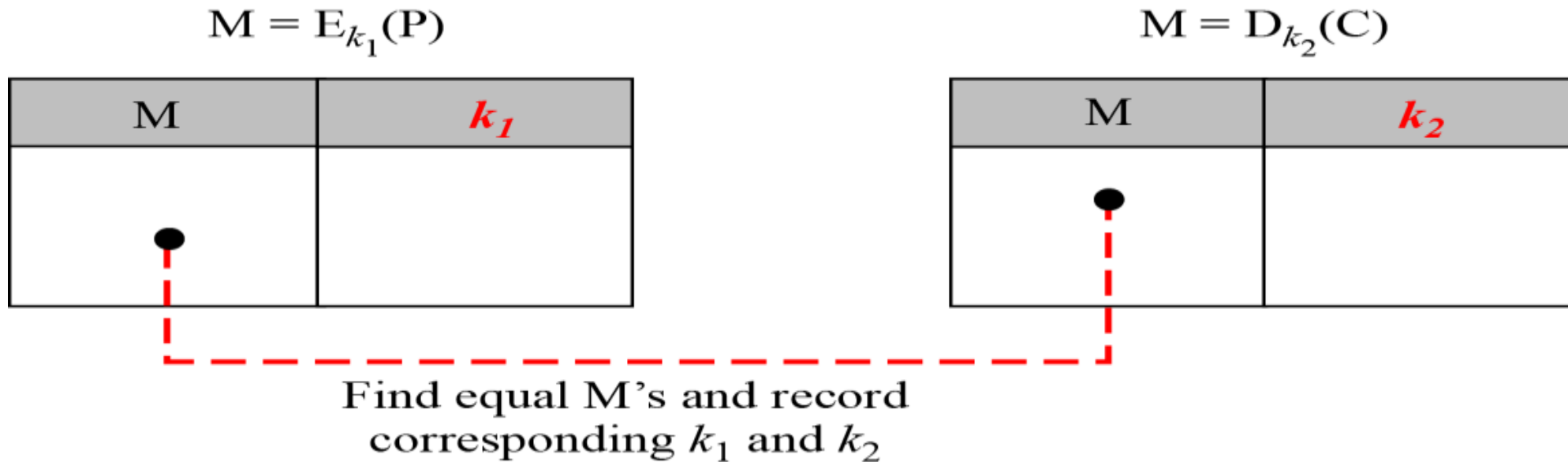
MEET-IN-THE-MIDDLE ATTACK

- ❑ The middle text, the text created by the first encryption or the first decryption, M , should be same

$$M = E_{K_1}(P) \quad M = D_{K_2}(C)$$

- ❑ Encrypt P using all possible values of K_1 and records all values obtained for M .
- ❑ Decrypt C using all possible values of K_2 and records all values obtained for M .
- ❑ Create **two tables sorted by M values.**
- ❑ Now compares the values for M until we find those pairs of K_1 & K_2 for which the value of **M is same in both tables.**

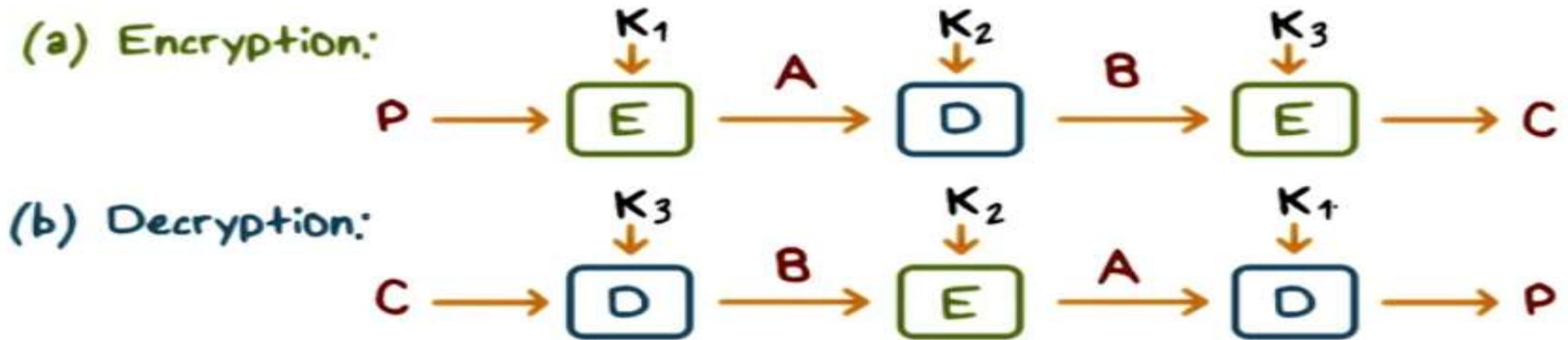
MEET-IN-THE-MIDDLE ATTACK



- ❑ Instead of using 2^{112} key search tests, we have to use 2^{56} key search tests two times.
- ❑ Moving from a Single DES to Double DES, we have to increased the strength from 2^{56} to 2^{57} .

Triple DES

- Triple DES (3DES) is a type of **encryption algorithm** that offers enhanced security through its **triple-layered encryption technique**.
- Triple Layer Encryption: 3DES uses **three different keys** to encrypt the plaintext three times, hence the name **Triple DES**.
- Variable Key Size: 3DES supports variable key sizes, ranging from 128 to 192 bits, offering enhanced security compared to DES.



Cont..

□ Encryption Process

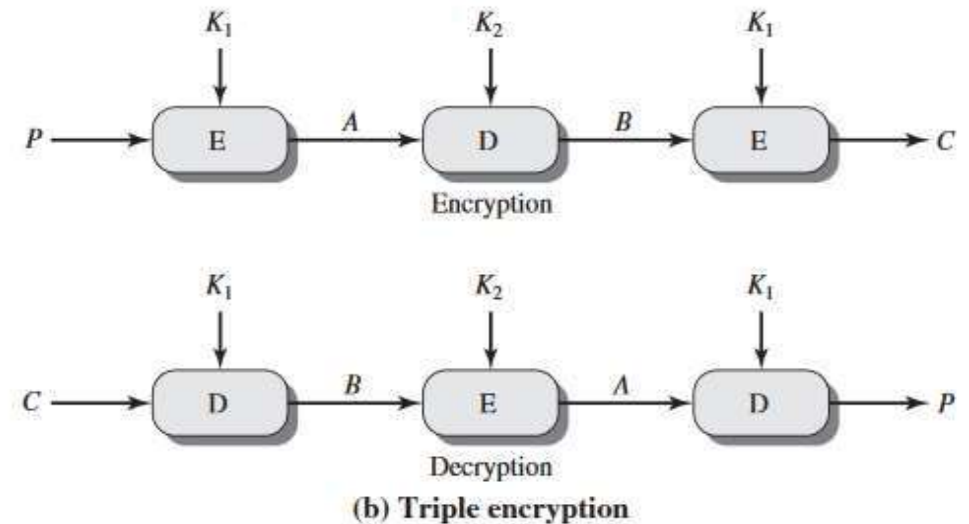
- **Key Generation:** Three unique keys are generated using a key derivation algorithm.
- **Initial Permutation:** The 64-bit plaintext is subjected to an initial permutation.
- **Three Rounds of Encryption:** The plaintext is encrypted three times, each time using a different key, to create three layers of encryption.
- **Final Permutation:** After the three rounds of encryption, a final permutation is applied to the output to produce the ciphertext.

□ Decryption Process

- The decryption process of 3DES is simply the reverse of the encryption process, with the ciphertext being fed into the algorithm and the steps being performed in reverse order, using the three keys in reverse order.

TRIPLE DES WITH 2-KEY

- ❑ Use three stages of DES for encryption and decryption.
- ❑ The 1st, 3rd stage use K_1 key and 2nd stage use K_2 key.
- ❑ To make triple DES compatible with single DES, the middle stage uses decryption in the encryption side and encryption in the decryption side.
- ❑ It's much stronger than double DES.



TRIPLE DES WITH 2-KEY

- ❑ The function follows an encrypt-decrypt-encrypt (EDE) sequence.

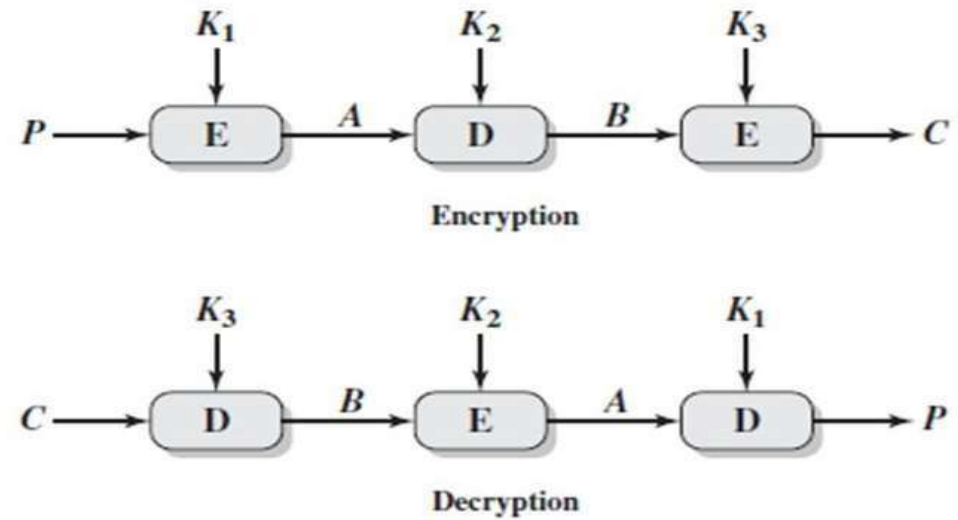
$$C = E(K_1, D(K_2, E(K_1, P)))$$

$$P = D(K_1, E(K_2, D(K_1, C)))$$

- ❑ By the use of triple DES with 2-key encryption, it raises the cost of **meet-in-the-middle attack** to 2^{112} .
- ❑ It has the **drawback of requiring a key length** of $56 \times 3 = 168\text{bits}$ which may be somewhat unwieldy.

TRIPLE DES WITH 3-KEY

- $C = E(K_3, D(K_2, E(K_1, P)))$
- $P = D(K_1, E(K_2, D(K_3, C)))$



Applications of 3DES

- **Financial Transactions:** online banking, credit card processing, and electronic fund transfers.
- **VPNs:** to provide secure communication between remote locations.
- **Healthcare Systems:** electronic health records and medical imaging systems.
- **Government Communications:** military communications and secure data transfers.

WEAK KEYS IN DES ALGORITHMS

- weak key
- in cryptography, a weak key is a key used **with a specific cipher**, makes the cipher behave in **some undesirable way**.
- To encrypt a message with weak keys are very unlikely to give **rise to a security problem**.
- It is considered desirable for a cipher to have **no weak keys**.

Weak key in DES

- A weak key is the one which **after parity drop operation**, consists either of all 0's, all 1's or half 0's and half 1's
- DES has 4 weak keys (64-bit)
 - 01010101 01010101
 - FEFEFEFE FEFEFEFE
 - E0E0E0E0 F1F1F1F1
 - 1F1F1F1F 0E0E0E0E
- Using weak keys, the **outcome of the Permuted Choice 1 (PC1)** in the DES key schedule leads to round keys (K 1---K16) being either all zeros, all ones or alternating zero-one patterns.
- Since all the **subkeys are identical**, and DES is a Feistel network,
- **encrypting twice with a weak key K produces** the original plaintext. – $E K(E K(x))=x$ for all x , i.e., the encryption and the decryption are the same.

Advanced Encryption Standard (AES)

Advanced Encryption Standard (AES)

- A **symmetric block cipher** which overcomes the **key size weakness of DES**.
- AES comes with the variable key sizes i.e. 128-bit key, 192-bit key and a 256-bit key.
- AES does not follow the **Feistel structure** in fact it operates on entire **plaintext**.
- AES can be applied **in both hardware and software**.
- AES operates on an **entire block of plain text** in the form of a **single matrix at each round which includes substitution and permutation**. AES has
 - 10 rounds -> 128-bit key
 - 12 rounds -> 192-bit key
 - 14 rounds -> 256-bit key
-

- AES operates on a 128-bit plain text block as a single **4X4 matrix** which would have a total size of **16 bytes**.
- Every **4 bytes would represent a word**.
- The 128-bit key is **expanded** to form an array containing **44, 32-bit words**. And at each round of AES 4 distinct words are served to the round key process from the expanded key.
- A round has four functions among which one is of permutation and three are of substitution: Substitute Byte, Mix Column and Add Round Key functions are **substitution** functions whereas Shift Rows is a **permutation** function.
- Both the **encryption** and **decryption** process starts with the **Add Round Key** function which is then followed by the **nine-round** which has all four substitutions and permutation function but the last tenth round has **only three functions**.
- Only the **Add Round key** function makes **use** of the **key**.
- During decryption, the **inverse functions** of Substitute Bytes, Mix Columns and Shift Rows are used and the inverse of Add Round key is carried out by XORing the same round key as in encryption with the cipher block.
- While decryption the sequence of keys used during encryption is reversed.
- In both the encryption and decryption, the **last round** always has only **three functions**, Mix Column is ignored at the last round of AES.
- Every intermediate result is stored in the 4X4 **state matrix**.

Plain Text
16 Bytes (128 bits)

Key
16 Bytes (128 bits)

Expand Key

Add Round Key

Substitute Bytes

Shift Rows

Mix Columns

Add Round Key

...

Substitute Bytes

Shift Rows

Mix Columns

Add Round Key

Substitute Bytes

Shift Rows

Add Round Key

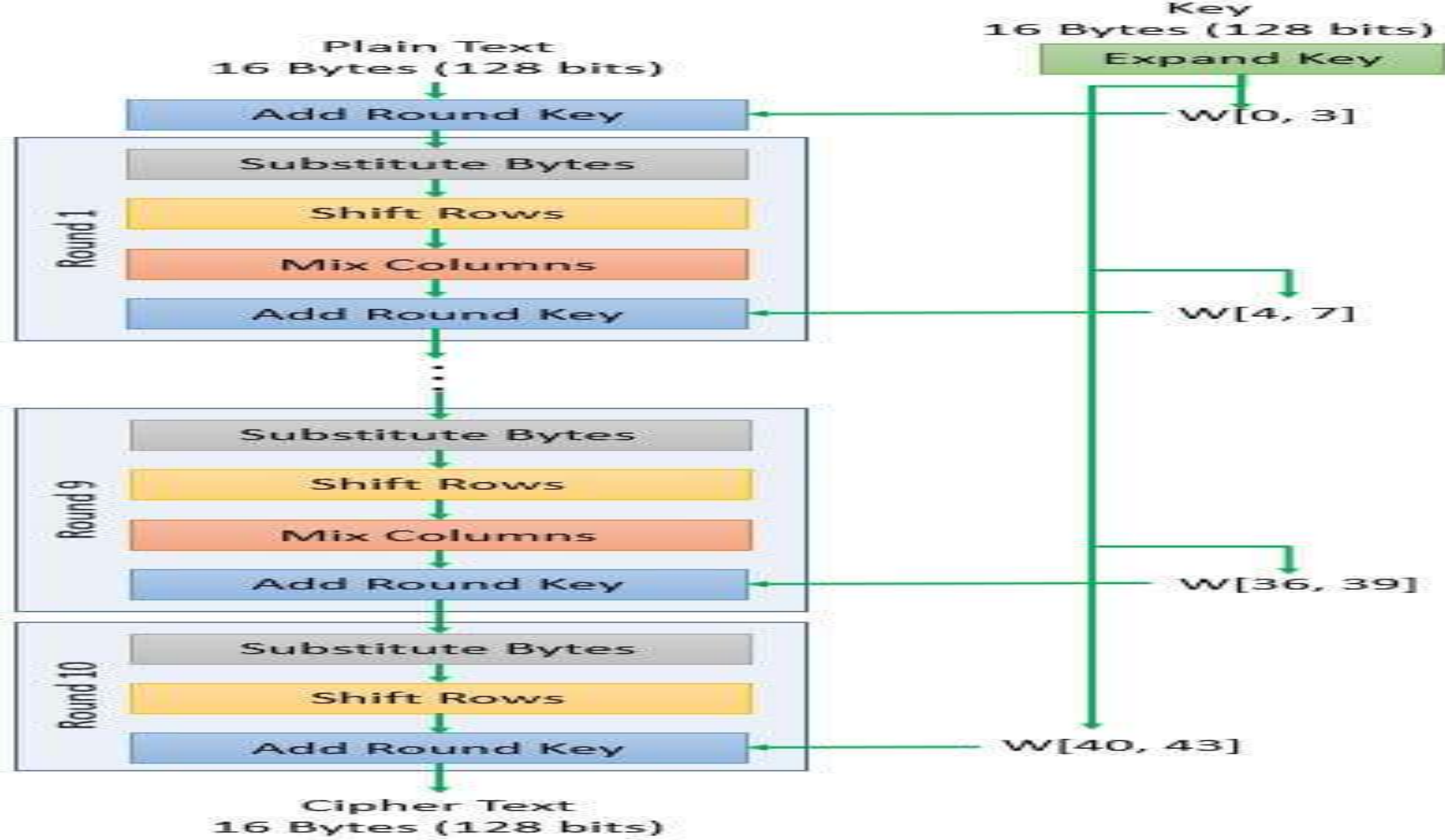
Cipher Text
16 Bytes (128 bits)

W[0, 3]

W[4, 7]

W[36, 39]

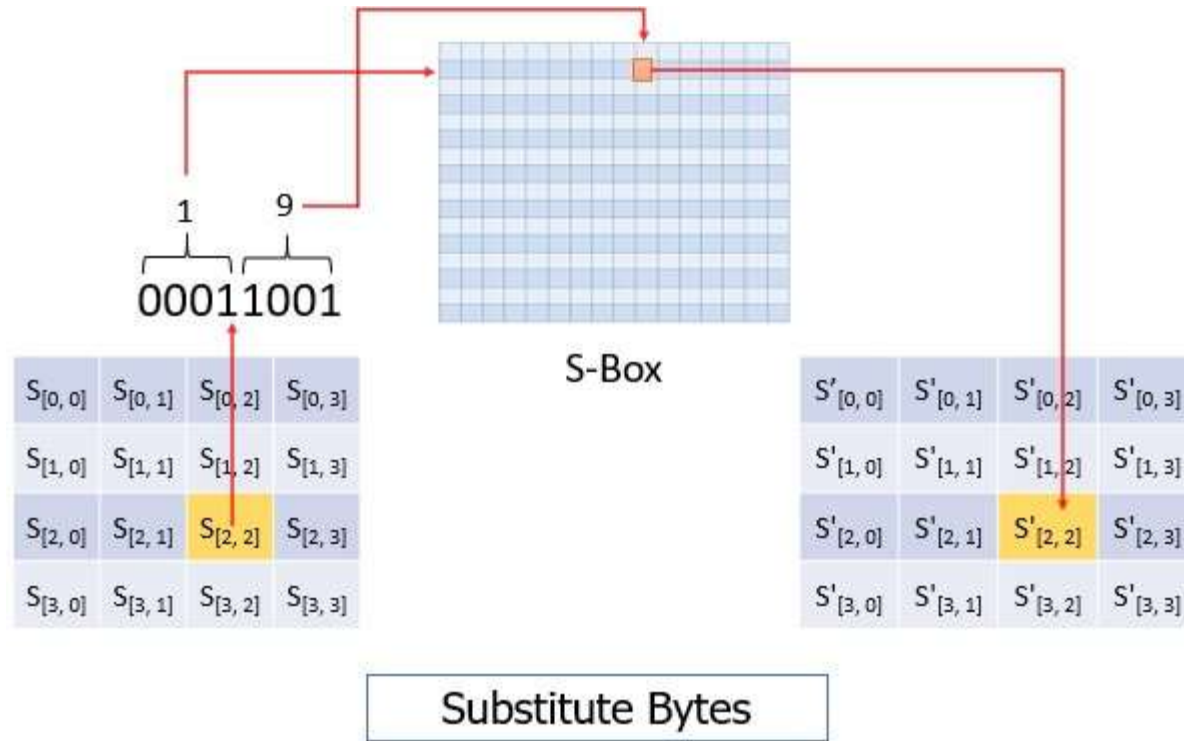
W[40, 43]



Advanced Encryption Standard (AES) Encryption

- **Step 1:** Initially the 16-byte key or a 4-word key is expanded to an array of 44 words where each word is of 4 bytes.
- **Step 2:** At the very first of encryption process the 16-byte plain text block or **4-word plain text block** is **XORed** with the **4-word key** i.e. W_0, W_1, W_2, W_3 . The resultant of this XOR is provided to the 1st round.
- **Step 3:** In the first round, the result of XOR is processed by the Substitute Bytes, Shift Rows, Mix Column and Add Round Key functions in the respective sequence.
- To the Add Round Key function, the next 4 words from the expanded key are provided that are W_4, W_5, W_6, W_7 . The corresponding result of first-round is provided to the second round.
- **Step 4:** All round till round nine, performs the same functions and in each round, a distinct key is provided from the expanded key. In round ten only three functions are performed on the input provided by round 9 that are Substitute Bytes, Shift Rows and Add Round Key function.
The result of round 10 is the **cipher text block of the corresponding plain text block**.

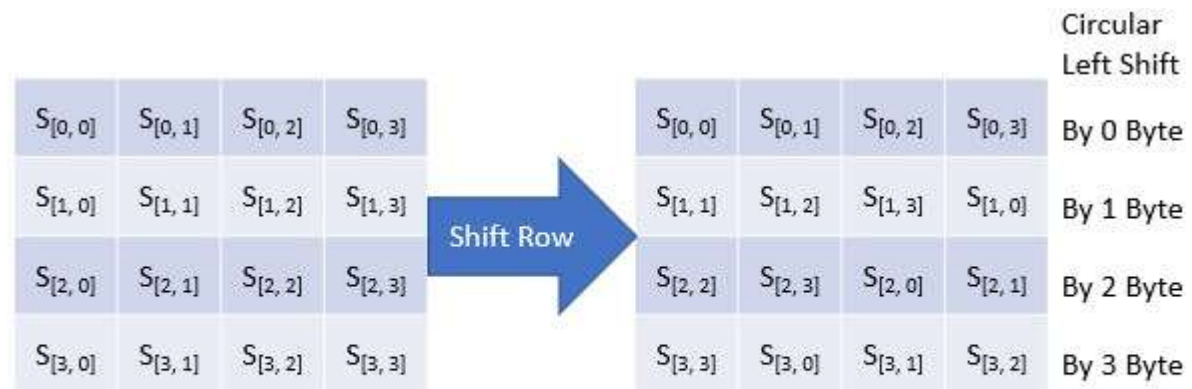
- **Round Functions**
- Each round function has four functions those are as follows:
- **Substitute Bytes**
- The **input** to Substitute Byte is a **4X4 state matrix** of 16 bytes where each element of the matrix is of 1 byte. Now AES has defined a **16X16 matrix** namely **S-box** which contains a permutation of 256 8-bit values.
- The Substitute Bytes function maps each byte element of state matrix to the new value using the following procedure.



- **tep 1: A byte element** in the state matrix would have 8-bits. The **leftmost 4-bit** is used to retrieve the **row value** of S-box and the **rightmost 4-bit** are used to retrieve the **column value** of S-box.
- **Step 2:** This row value and column value act as an **index** to get the new value from the **S Box**.
- The Substitute Byte function maps each element of 4X4 state matrix to the new value and forwards this newly formed 4X4 State matrix to **Shift Rows function**.

- **Shift Rows**

- On the **first row**, the circular left shift is performed by **0 bytes**.
- On the **second row**, a circular left shift is performed by **1 byte**.
On the **third row**, the circular left shift is performed by **2 bytes**.
On the **fourth row** of the input state matrix, the circular left shift is performed by **3 bytes**.
- The resultant 4X4 state matrix of Shift Rows function is forwarded to the Mix Column function.



Shift Rows

- **Mix Columns**

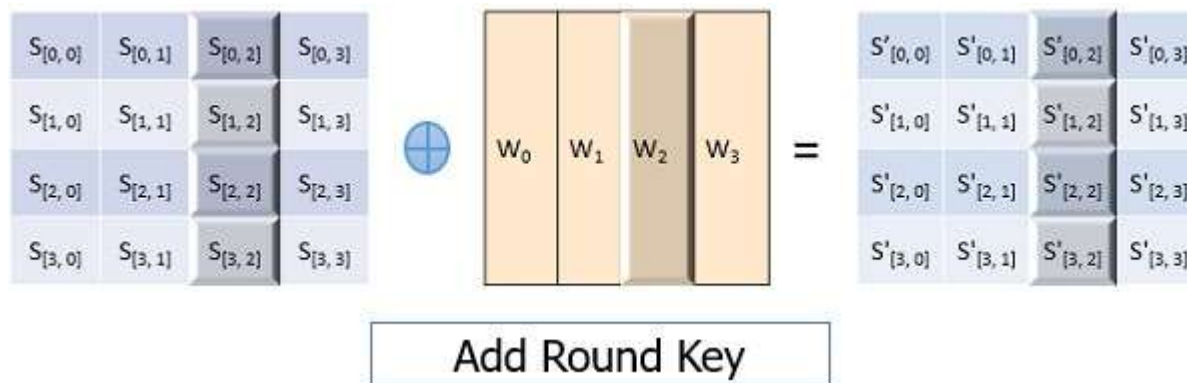
- The input 4X4 state matrix is **multiplied** with a **constant predefined matrix** as you can see in the figure below:
- Each byte element of the resultant matrix of Mix Column function is **the sum of the product of one row of the defined matrix and one column of state matrix**. The product matrix of the Mix Column function is forwarded to the last function of a round i.e. **Add Round Key**.

•

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \times \begin{bmatrix} S_{[0,0]} & S_{[0,1]} & S_{[0,2]} & S_{[0,3]} \\ S_{[1,0]} & S_{[1,1]} & S_{[1,2]} & S_{[1,3]} \\ S_{[2,0]} & S_{[2,1]} & S_{[2,2]} & S_{[2,3]} \\ S_{[3,0]} & S_{[3,1]} & S_{[3,2]} & S_{[3,3]} \end{bmatrix} = \begin{bmatrix} S'_{[0,0]} & S'_{[0,1]} & S'_{[0,2]} & S'_{[0,3]} \\ S'_{[1,0]} & S'_{[1,1]} & S'_{[1,2]} & S'_{[1,3]} \\ S'_{[2,0]} & S'_{[2,1]} & S'_{[2,2]} & S'_{[2,3]} \\ S'_{[3,0]} & S'_{[3,1]} & S'_{[3,2]} & S'_{[3,3]} \end{bmatrix}$$

Mix Columns

- **Add Round Key**
- In the Add Round Key function, the input state matrix is **XORed** with the **4-words unique key**.
- In each round the key used to XOR with state matrix is **distinct**.
- Add Round Key function is a **column-wise** function, a 4-byte state matrix column is XORed with a 4-byte word of a key. It can also be taken as **byte-level** function.



- Advanced Encryption Standard (AES) Decryption
- Encryption and Decryption process are the same and starts with the Add Round Key Function.
- The 16-byte cipher text in the form of **4X4 state matrix** is XORed with the **unique 4-word key**.
- **The key sequence in encryption is reversed** during the decryption.
- Advantage and Disadvantage
- **Advantages**
 - AES can be implemented on both **hardware and software**.
 - AES has three key length **128-bits, 192-bits and 256-bits**.
 - AES is implemented in a wide range of application as it is defined as the standard by NIST.
- **Disadvantage**
 - The key used in AES if not employed properly it can cause a cryptanalytic attack. Therefore, key scheduling should be done carefully.

THANK YOU?.....

E-links

- <https://www.101computing.net/symmetric-vs-asymmetric-encryption/>
- <https://www.101computing.net/caesar-cipher/>
- <https://www.101computing.net/mono-alphabetic-substitution-cipher/>

